Hodoscope Paddle Efficiencies

Comparison: RS67 vs RS57-70 Final

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Input Data

Input files used:

- `effies_rs57.xlsx`
- `effies_rs59.xlsx`
- `effies_rs62.xlsx`
- `effies_rs67.xlsx`
- `effies_rs70.xlsx`

Hodoscope efficiency tables used:

- `hodoscope_eff_RS67_final_2026-01-31.tsv` (previous paddle efficiency table)
- `hodoscope_eff_RS57-70_final.tsv` (latest paddle efficiency table)

Note

The current study is only focused on runs 2 and 3 only. We will do the same study for runs 5 and 6 later as a separate study.

Analysis Procedure

- **Average Efficiency:** Calculated the arithmetic mean of efficiency across the runs.
- **Propagated Error:** Errors were combined in quadrature:

$$\sigma_{prop} = \frac{1}{N}\sqrt{\sum_{i=1}^{N}\sigma_i^2}$$

- **Standard Deviation (σ_{sample}):** Calculated the sample standard deviation of the efficiency distribution to determine the spread:

$$\sigma_{sample} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(\epsilon_i - \bar{\epsilon})^2}$$

- **Error Bar Substitution:** If the standard deviation is larger than the maximum propagated error, both lower and upper error bars are replaced by it:

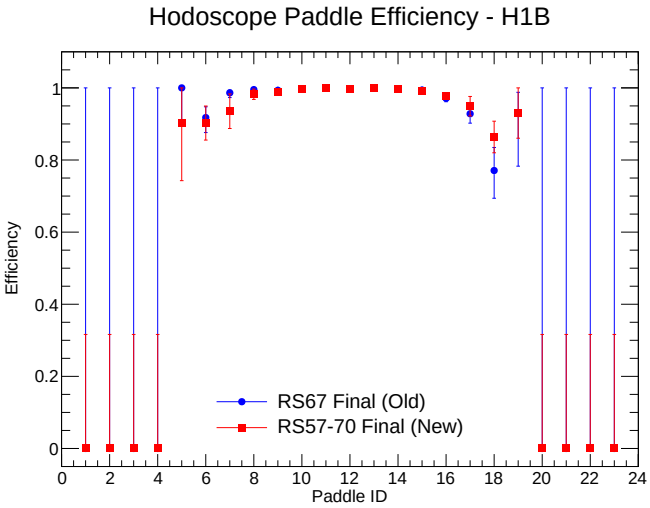
$$\text{If } \sigma_{sample} > \max(\sigma_{prop}^{up}, \sigma_{prop}^{down}) \implies \sigma^{up,down} = \sigma_{sample}$$

- **Physical Limit Capping:** To respect physical boundaries, if the average efficiency plus

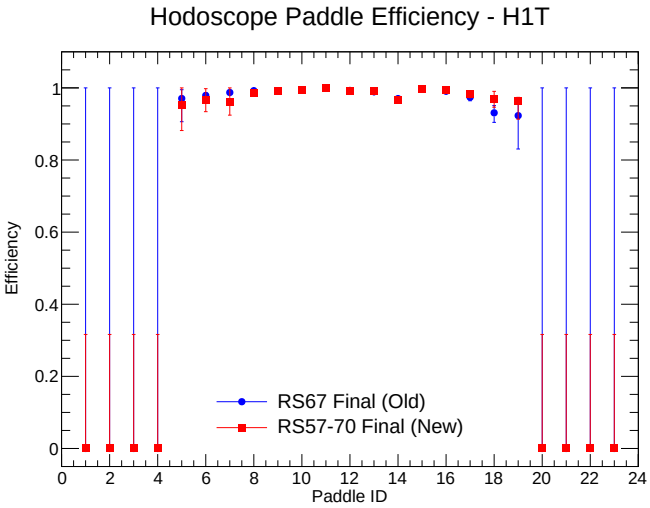
Paddles with Upper Error Bar Capped at 1.0 Bound

Plane	Paddle ID	Original σ_{up}	Capped σ_{up}
H1B	5	0.1592	0.0981
H1B	19	0.0707	0.0690
H1T	5	0.0704	0.0479
H1T	7	0.0378	0.0378
H2B	2	0.5164	0.4000
H2B	8	0.0006	0.0005
H2T	8	0.0004	0.0004
H2T	9	0.0004	0.0004
H3B	4	0.0086	0.0075
H3B	16	0.0318	0.0182
H3T	8	0.0011	0.0004

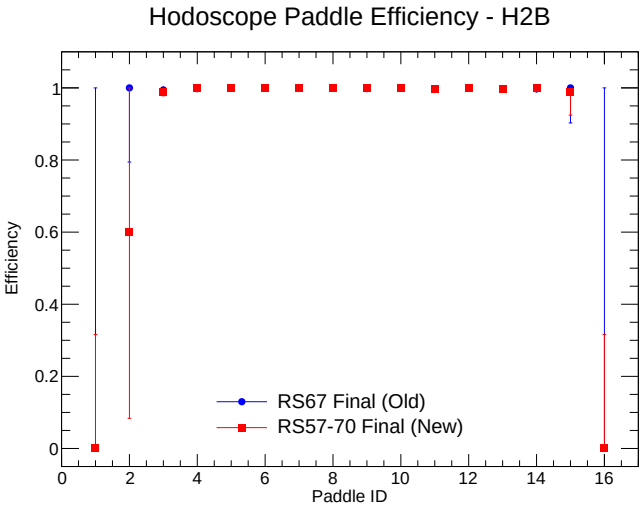
Hodoscope Efficiency: H1B



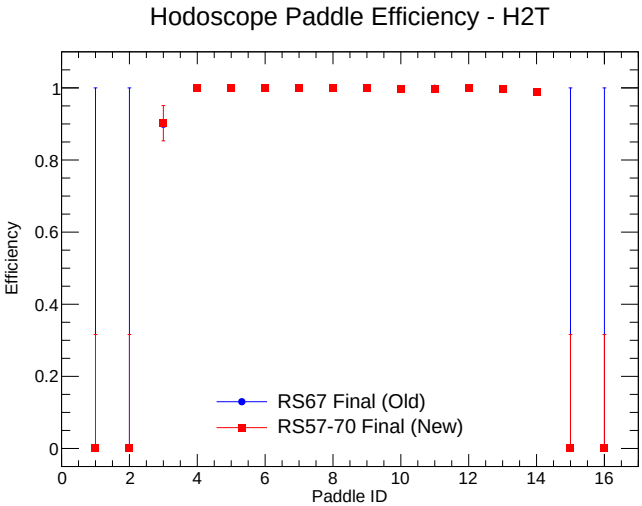
Hodoscope Efficiency: H1T



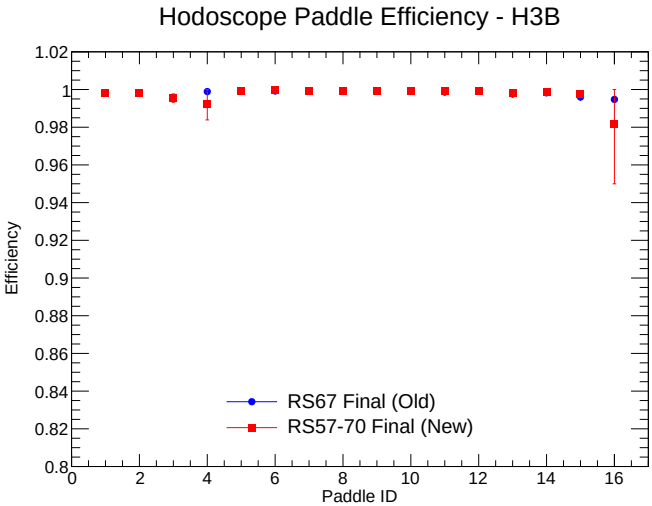
Hodoscope Efficiency: H2B



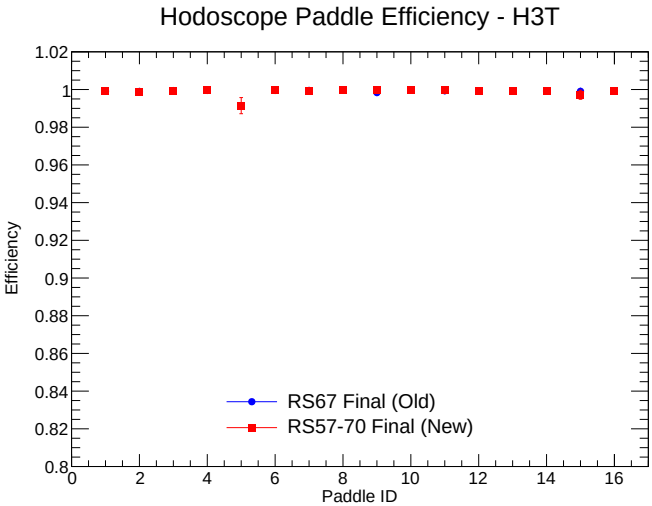
Hodoscope Efficiency: H2T



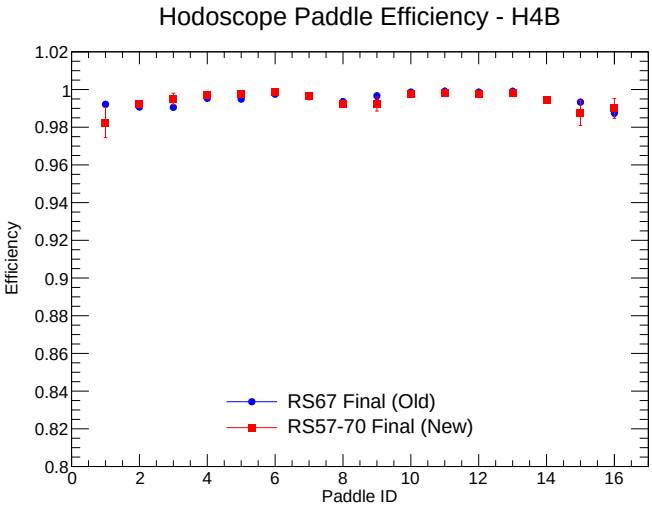
Hodoscope Efficiency: H3B



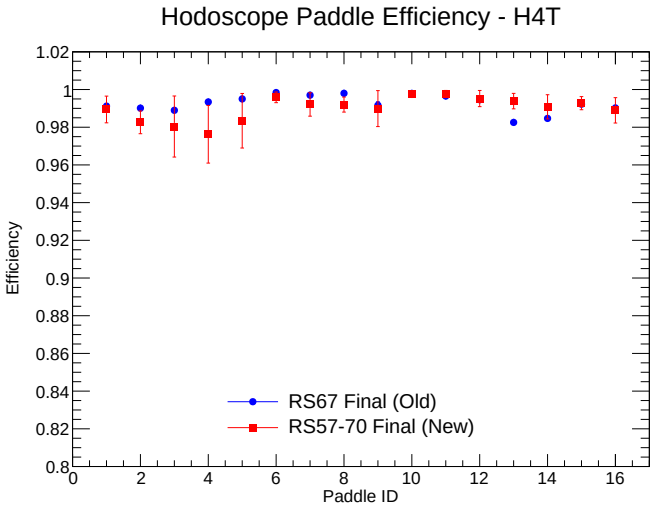
Hodoscope Efficiency: H3T



Hodoscope Efficiency: H4B



Hodoscope Efficiency: H4T

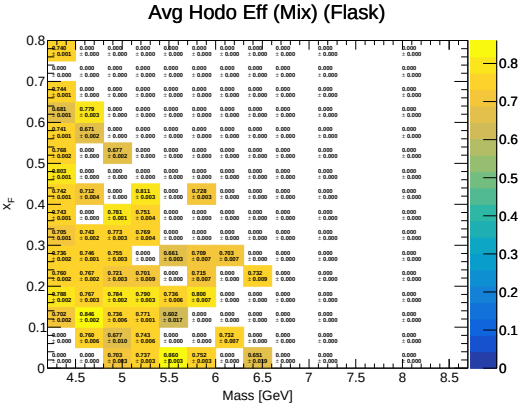


Hodoscope Efficiency Comparison

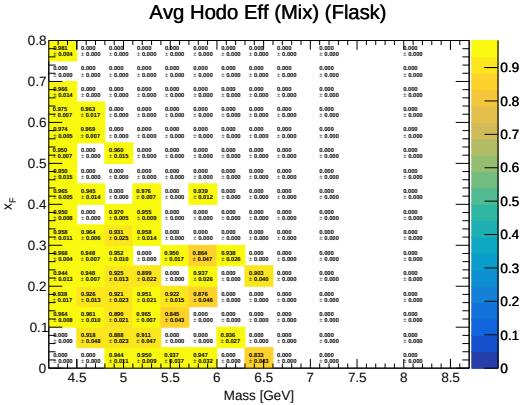
(Previous Vs Latest)

Comparison: E_mix_hodo_Flask

Previous (RS67)



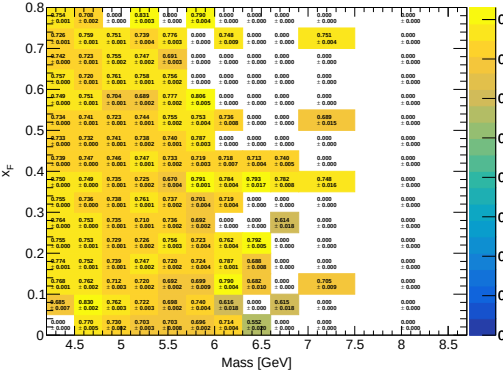
Latest (RS57-70)



Comparison: E_mix_hodo_LD2

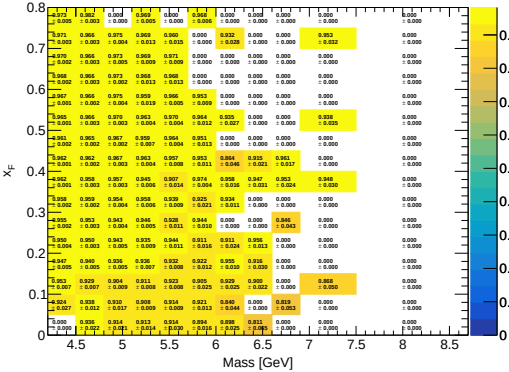
Previous (RS67)

Avg Hodo Eff (Mix) (LD2)



Latest (RS57-70)

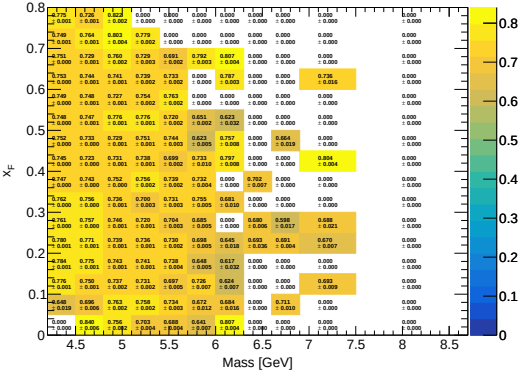
Avg Hodo Eff (Mix) (LD2)



Comparison: E_mix_hodo_LH2

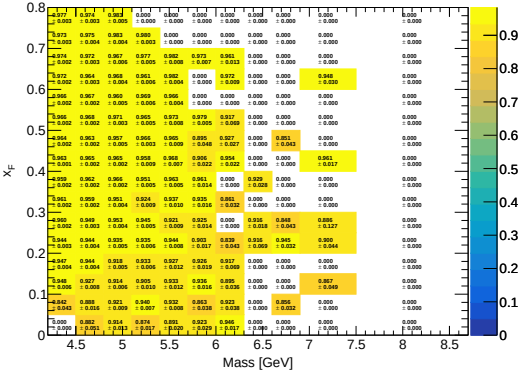
Previous (RS67)

Avg Hodo Eff (Mix) (LH2)



Latest (RS57-70)

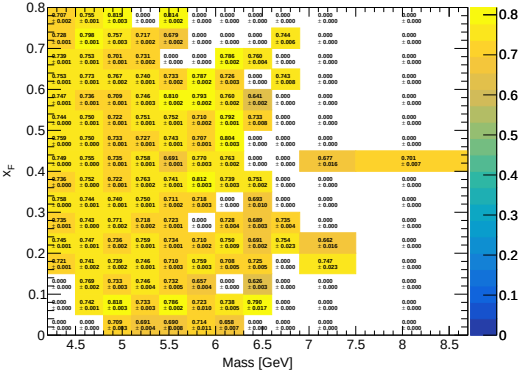
Avg Hodo Eff (Mix) (LH2)



Comparison: E_total_hodo_Flask

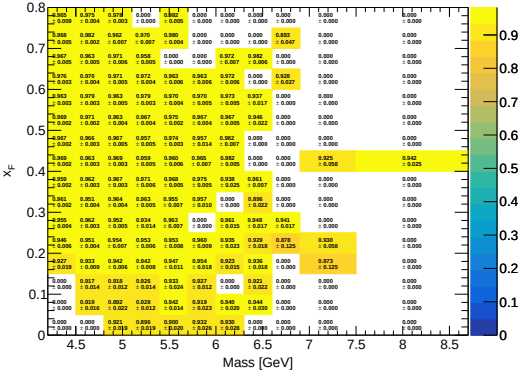
Previous (RS67)

Avg Hodo Eff (Total) (Flask)



Latest (RS57-70)

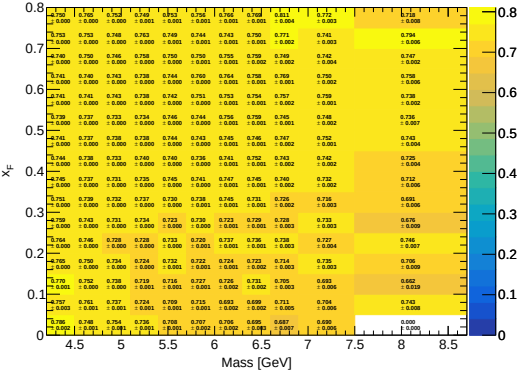
Avg Hodo Eff (Total) (Flask)



Comparison: E_total_hodo_LD2

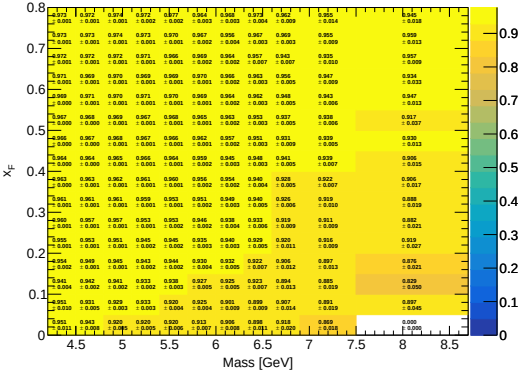
Previous (RS67)

Avg Hodo Eff (Total) (LD2)



Latest (RS57-70)

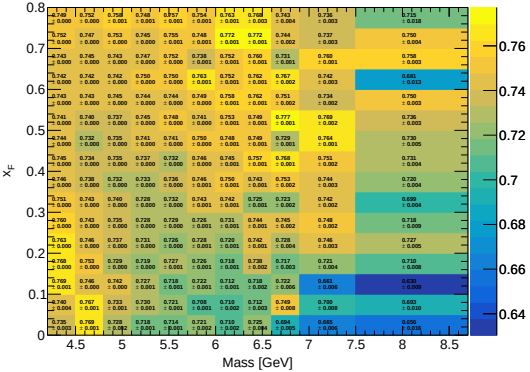
Avg Hodo Eff (Total) (LD2)



Comparison: E_total_hodo_LH2

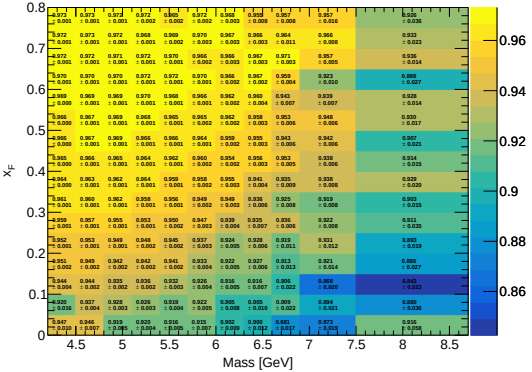
Previous (RS67)

Avg Hodo Eff (Total) (LH2)



Latest (RS57-70)

Avg Hodo Eff (Total) (LH2)



Questions

Questions & Discussion

1. **Can you confirm that your blue points are from one of Harsha's TSV files from (say) version 6 of his DocDB?**
   They correspond to DocDB 11467-v2. Harsha should be able to verify this. As noted previously, Harsha sent me the zip file directly on January 31st, which was helpful in addressing the feedback from my January 20th talk (DocDB 11460).

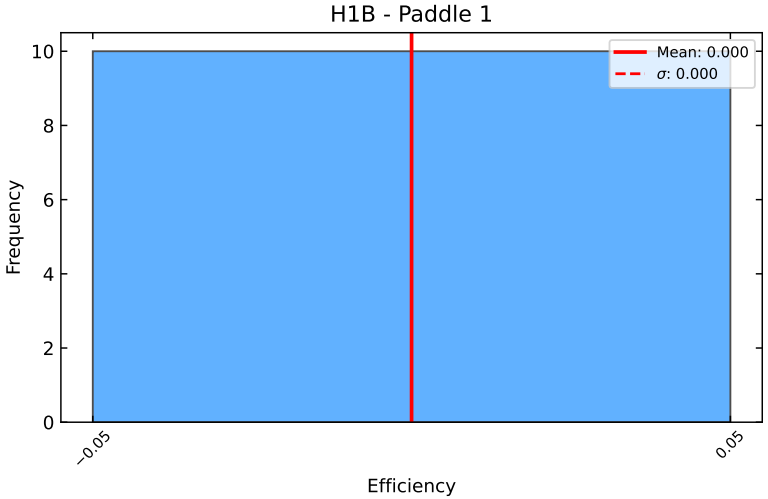
2. **Are you using RS67 data to produce the latest hodoscope paddle efficiency table?**
   Yes, I do. Please check the console output.

3. **Are there ROOT files that contain: [`result`, `result_mix`] TTrees made for roadsets: 57-70 like Abinash's files for RS67 (not the gridjob output that requires to reprocess)?**

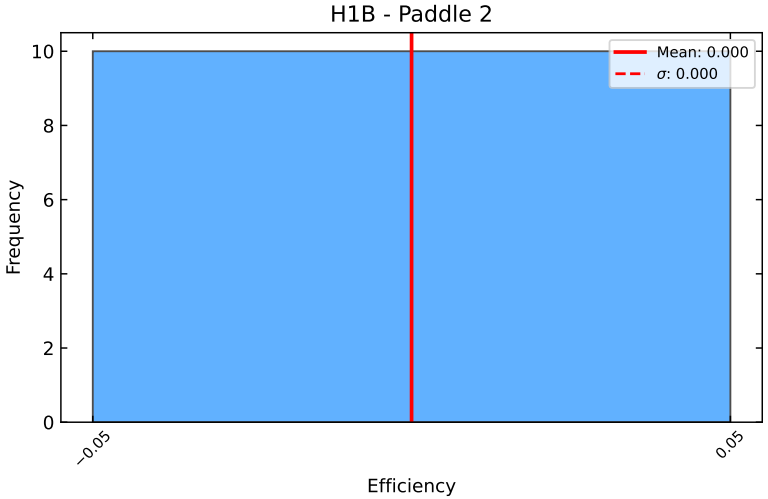
Appendix

Hodoscope Efficiency Distributions with Mean and RMS

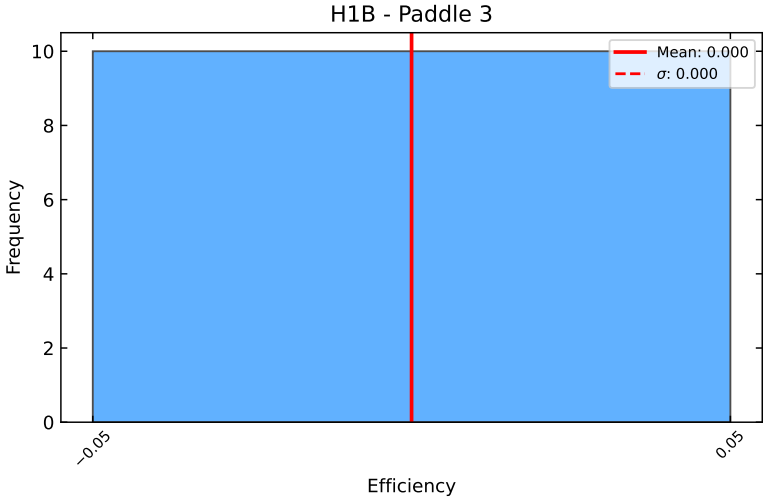
Efficiency Distribution: H1B - Paddle 1



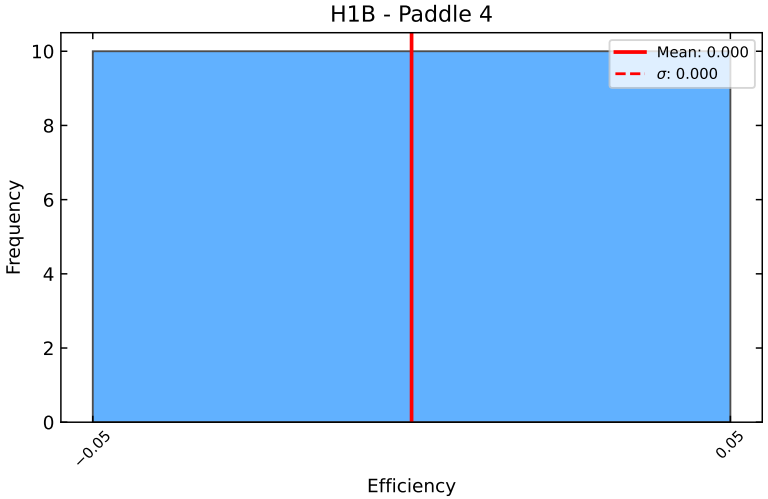
Efficiency Distribution: H1B - Paddle 2



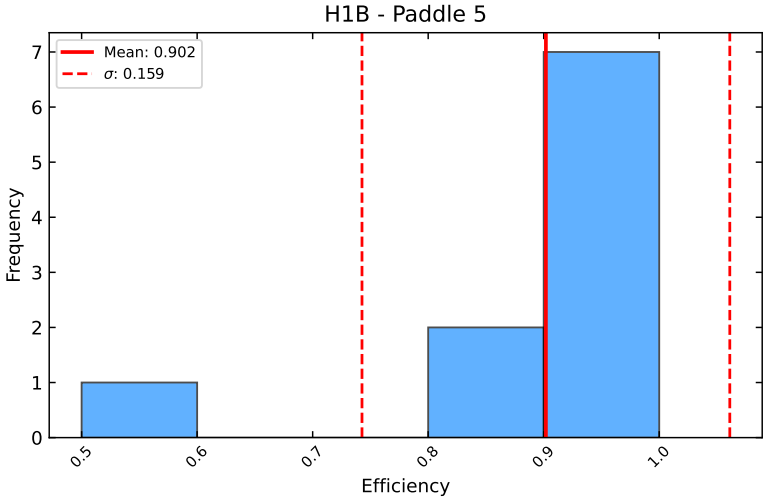
Efficiency Distribution: H1B - Paddle 3



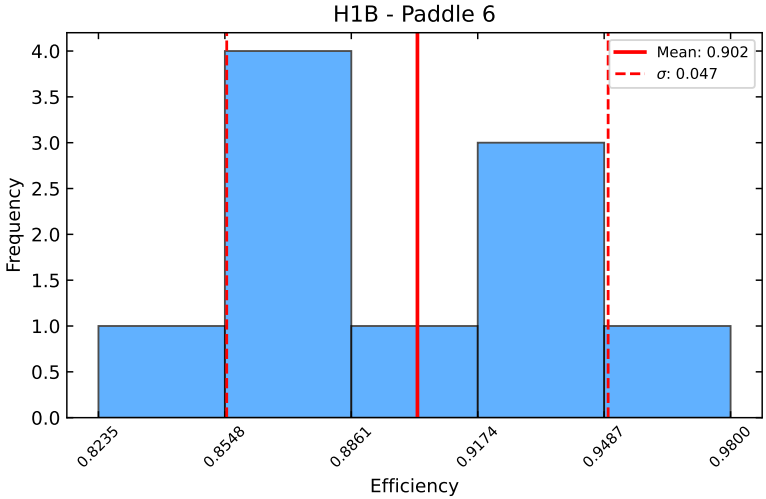
Efficiency Distribution: H1B - Paddle 4



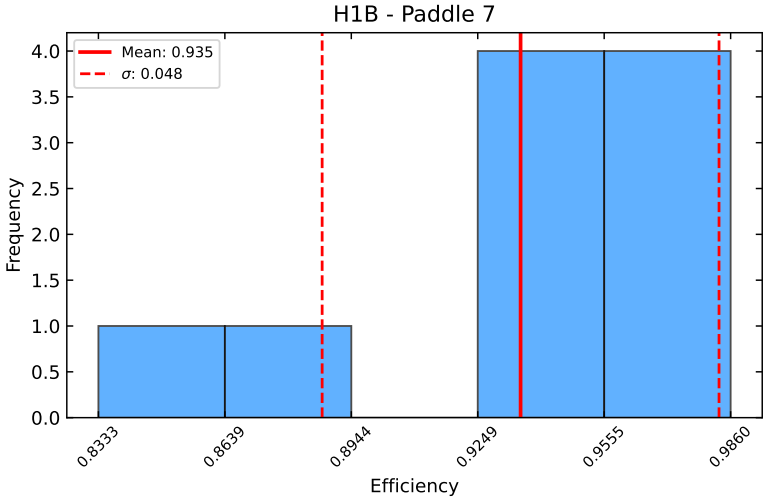
Efficiency Distribution: H1B - Paddle 5



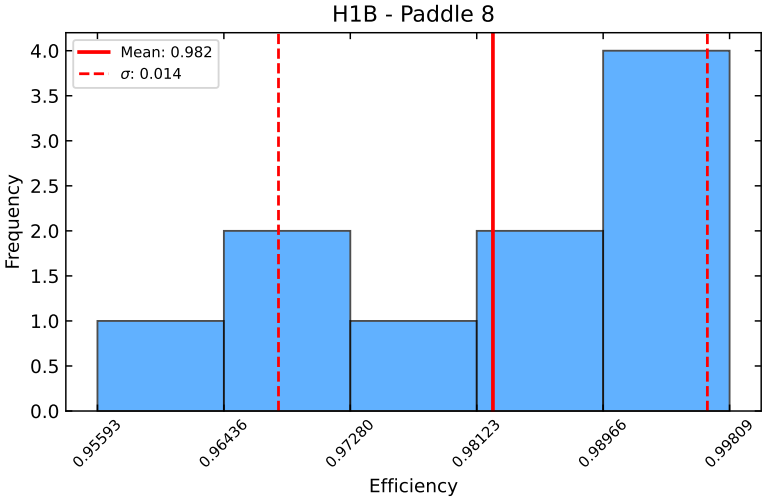
Efficiency Distribution: H1B - Paddle 6



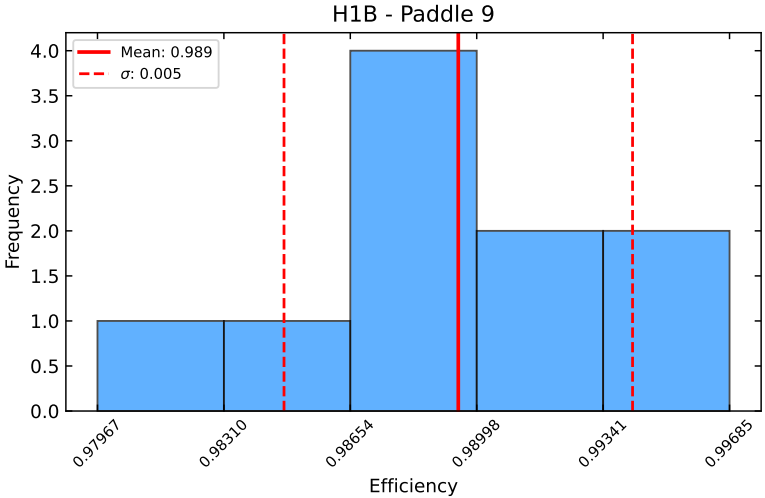
Efficiency Distribution: H1B - Paddle 7



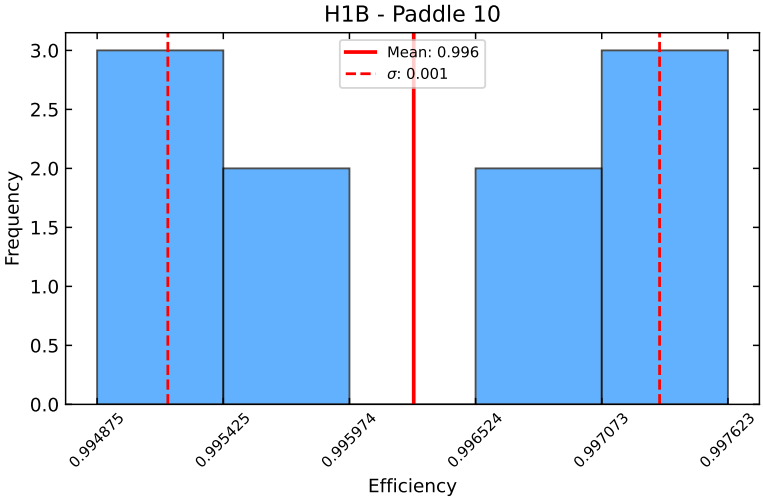
Efficiency Distribution: H1B - Paddle 8



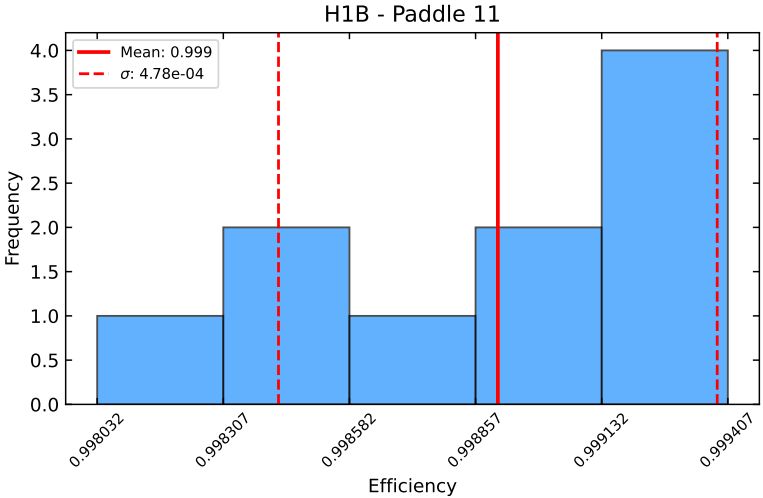
Efficiency Distribution: H1B - Paddle 9



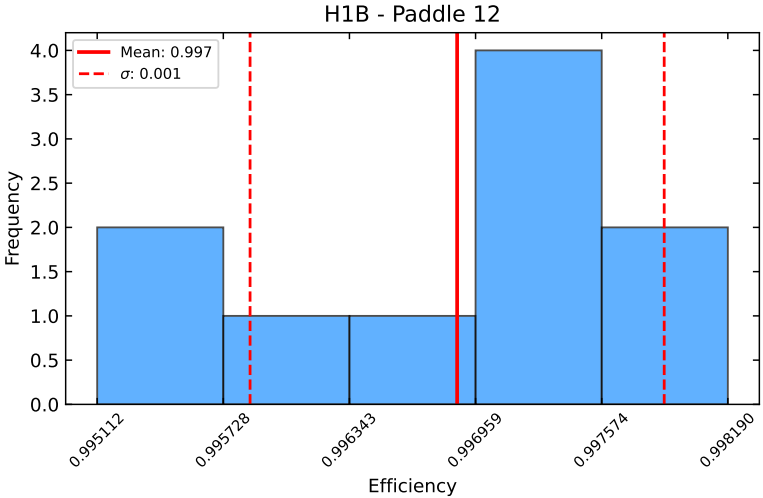
Efficiency Distribution: H1B - Paddle 10



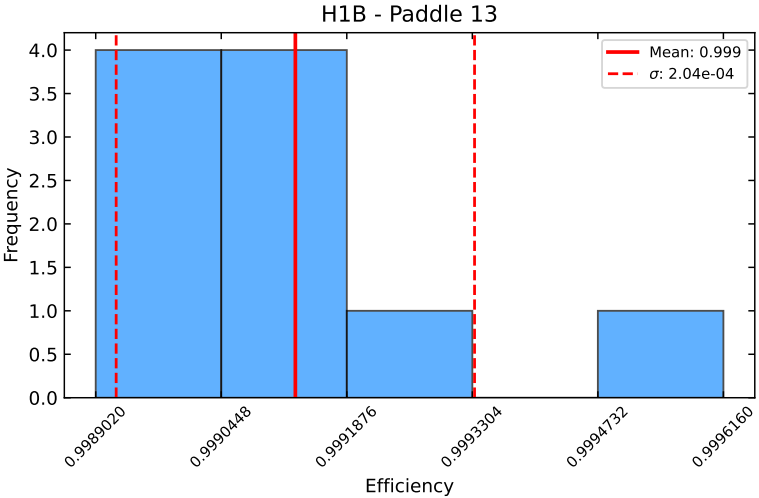
Efficiency Distribution: H1B - Paddle 11



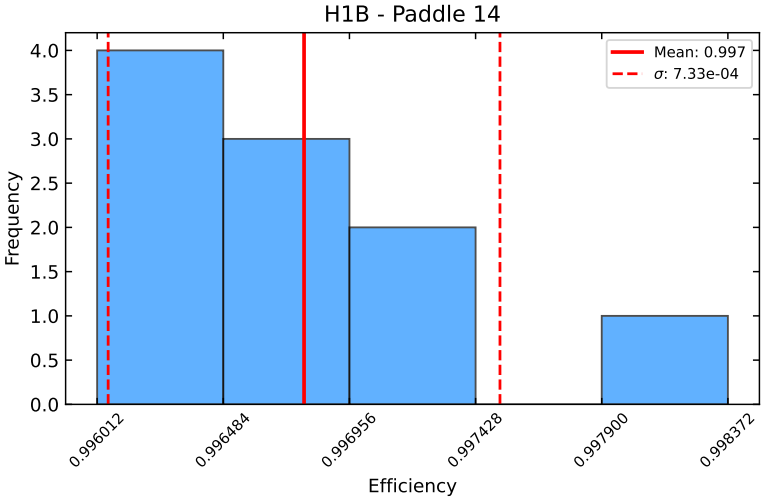
Efficiency Distribution: H1B - Paddle 12



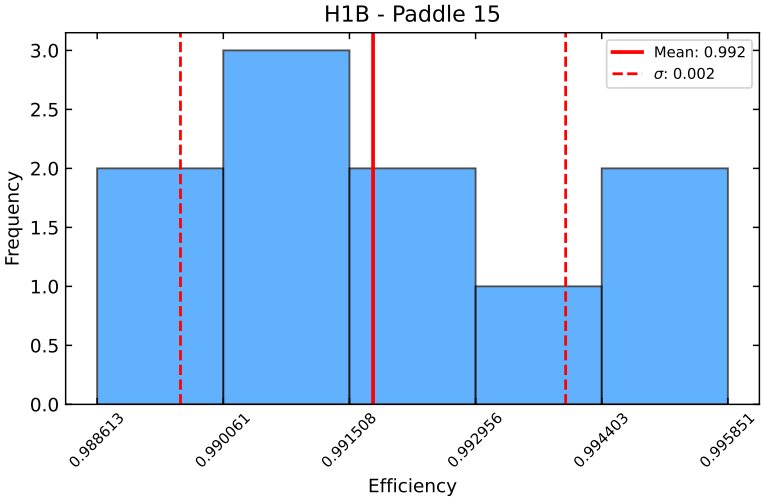
Efficiency Distribution: H1B - Paddle 13



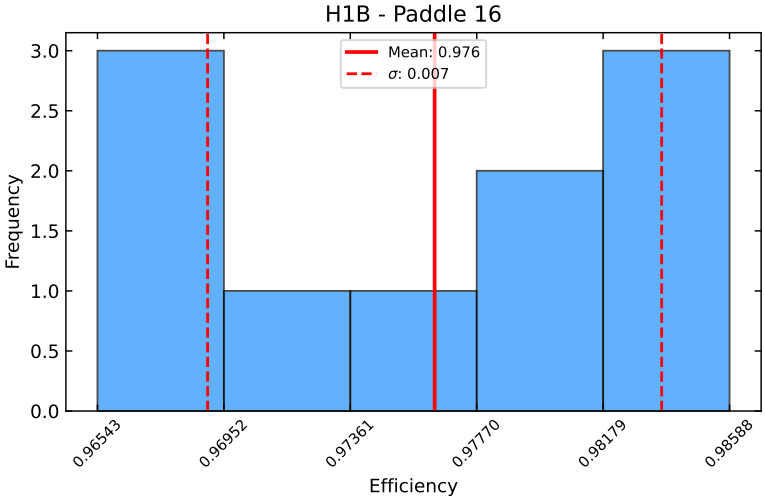
Efficiency Distribution: H1B - Paddle 14



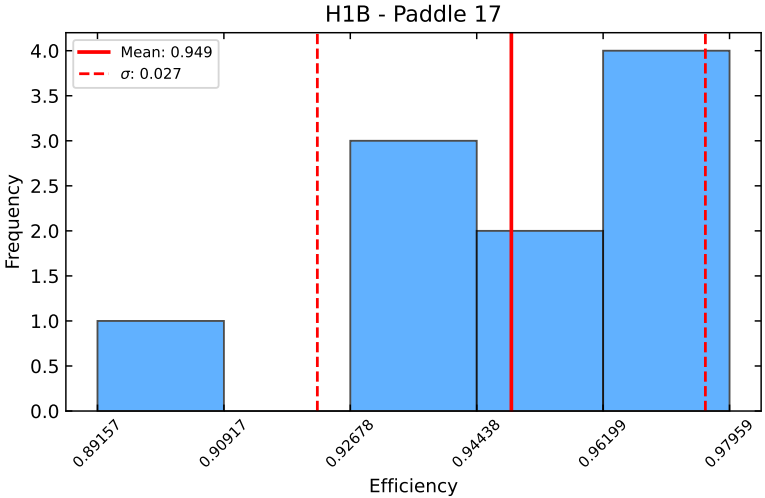
Efficiency Distribution: H1B - Paddle 15



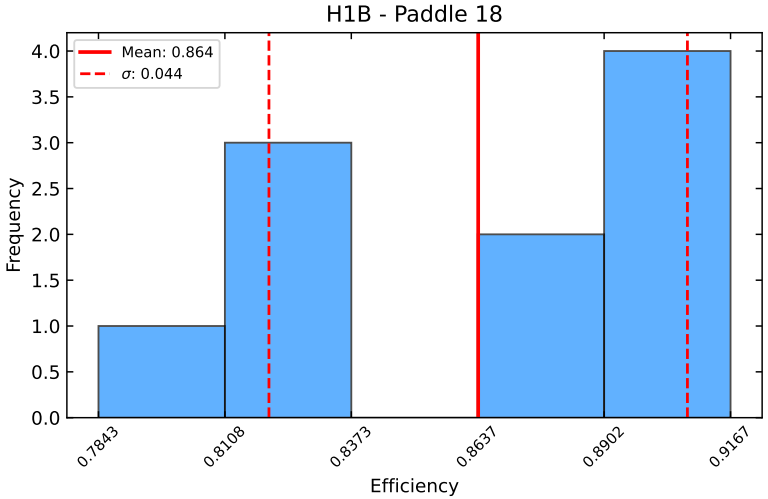
Efficiency Distribution: H1B - Paddle 16



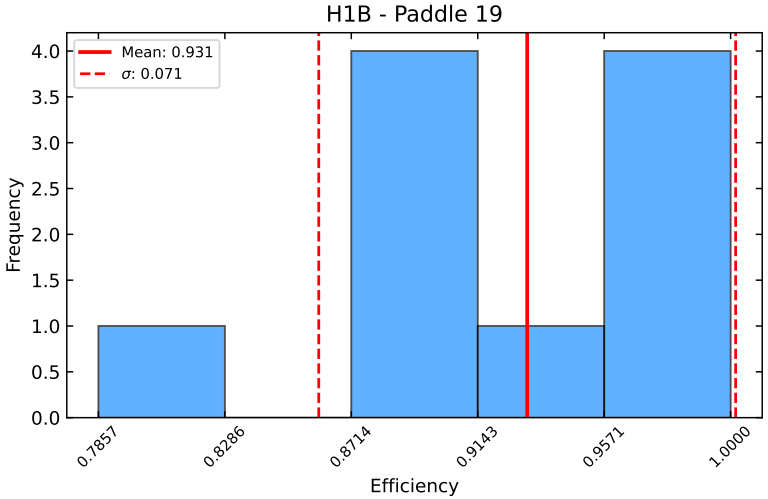
Efficiency Distribution: H1B - Paddle 17



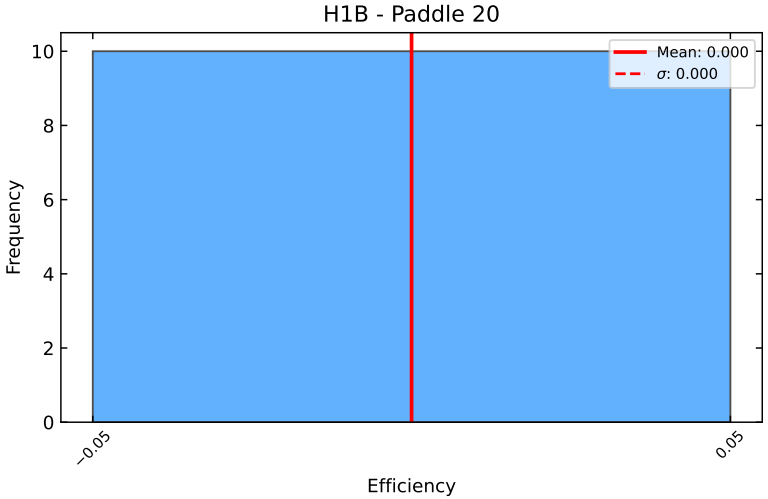
Efficiency Distribution: H1B - Paddle 18



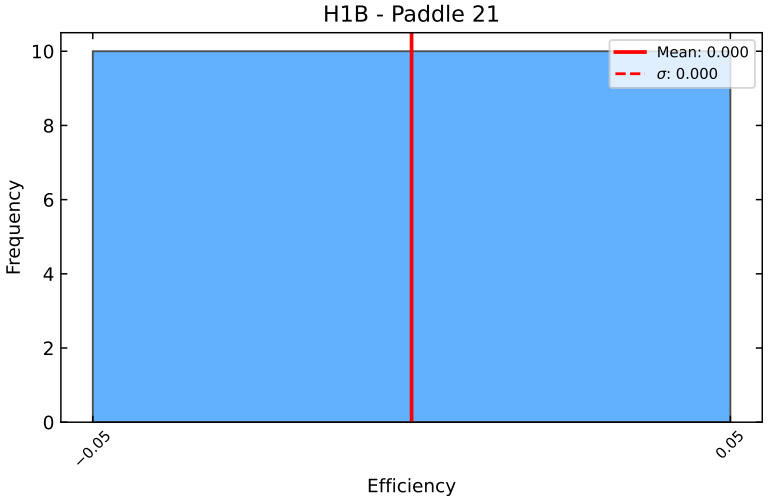
Efficiency Distribution: H1B - Paddle 19



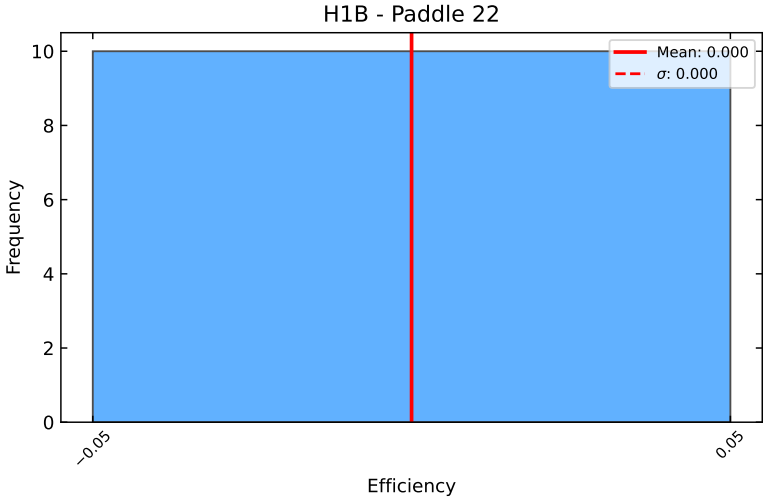
Efficiency Distribution: H1B - Paddle 20



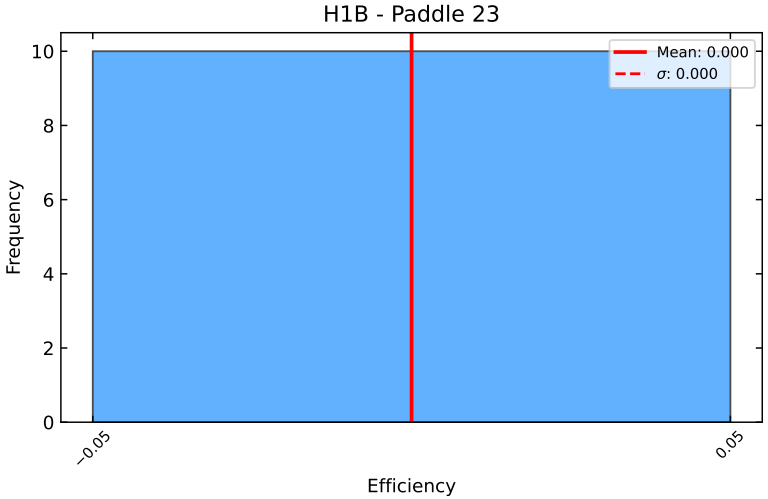
Efficiency Distribution: H1B - Paddle 21



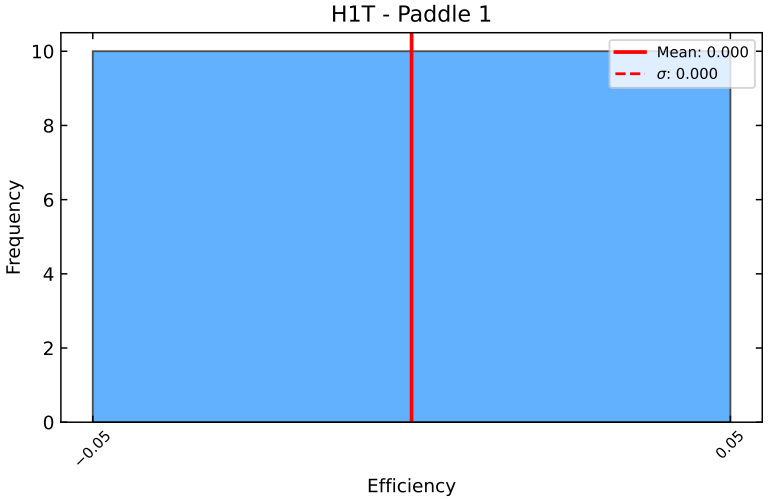
Efficiency Distribution: H1B - Paddle 22



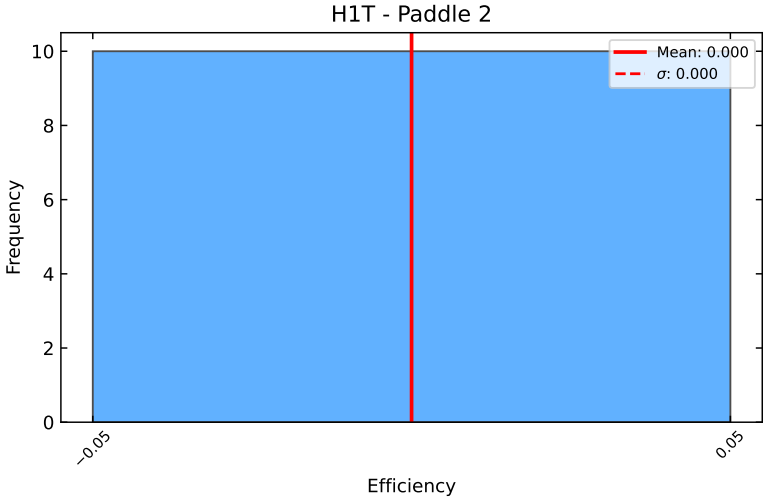
Efficiency Distribution: H1B - Paddle 23



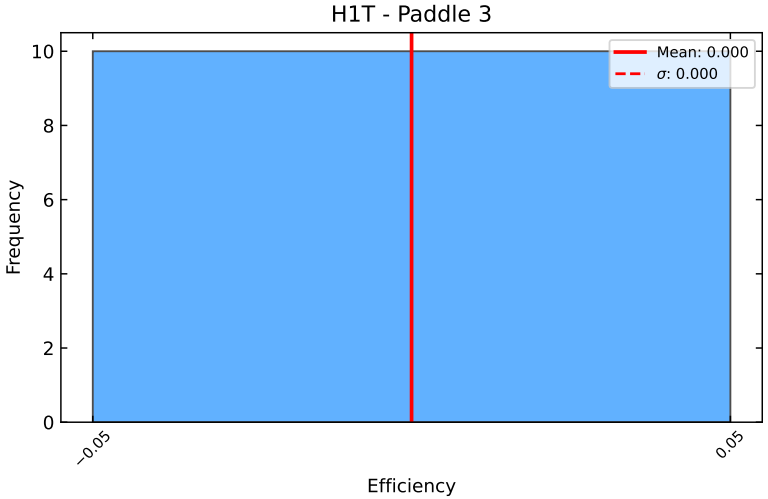
Efficiency Distribution: H1T - Paddle 1



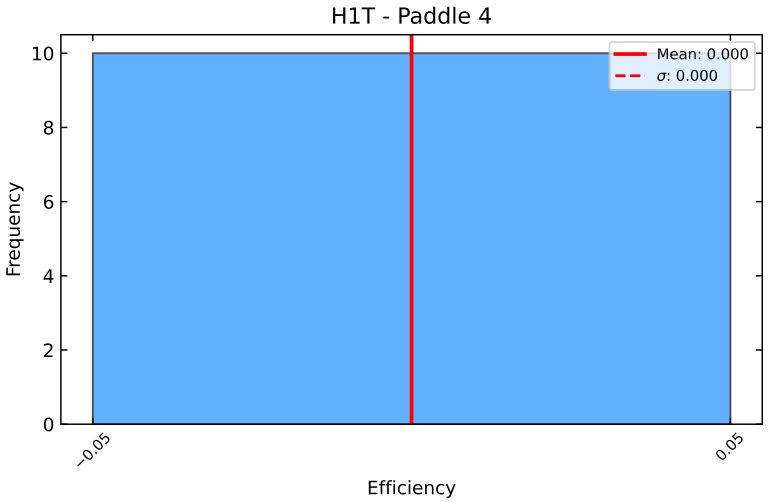
Efficiency Distribution: H1T - Paddle 2



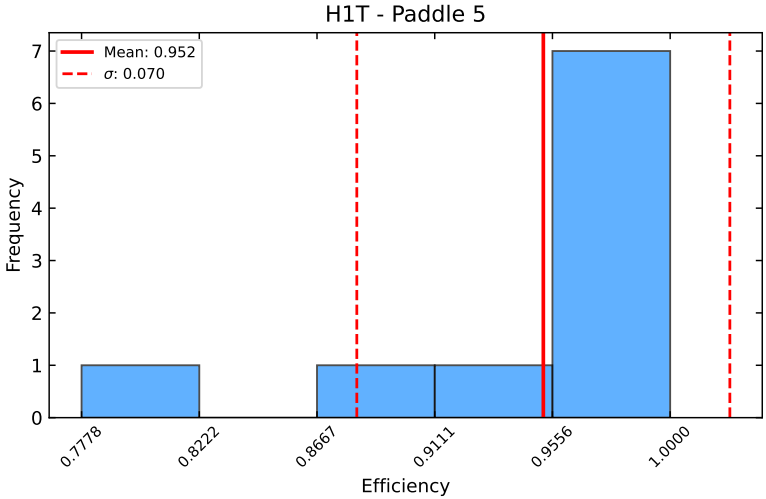
Efficiency Distribution: H1T - Paddle 3



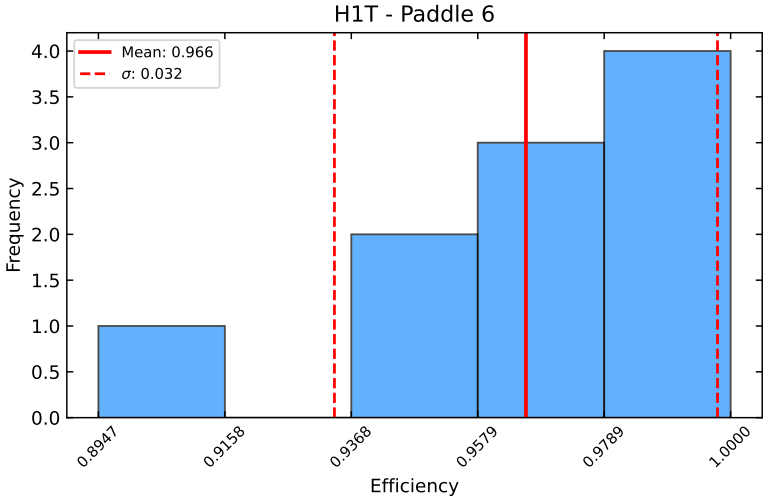
Efficiency Distribution: H1T - Paddle 4



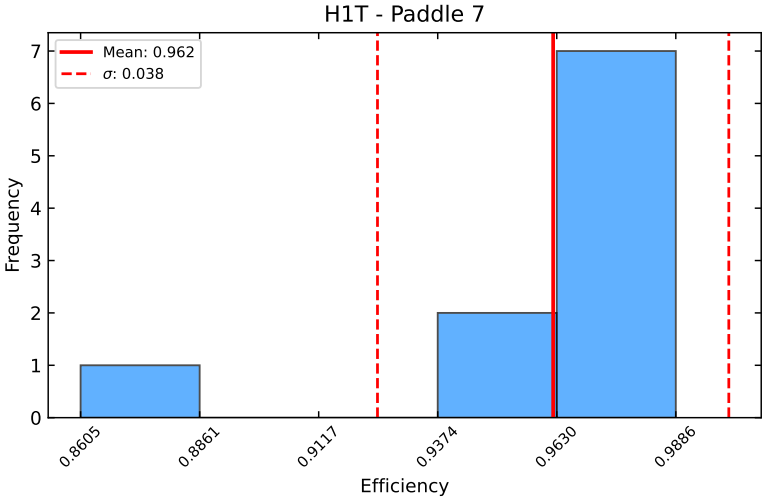
Efficiency Distribution: H1T - Paddle 5



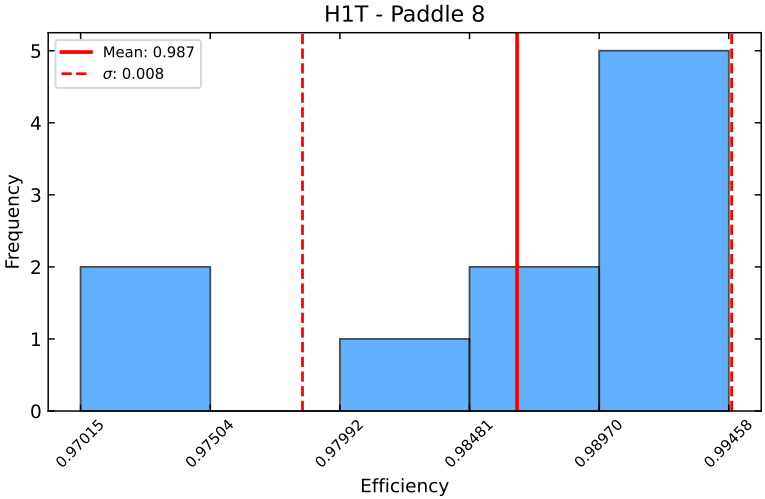
Efficiency Distribution: H1T - Paddle 6



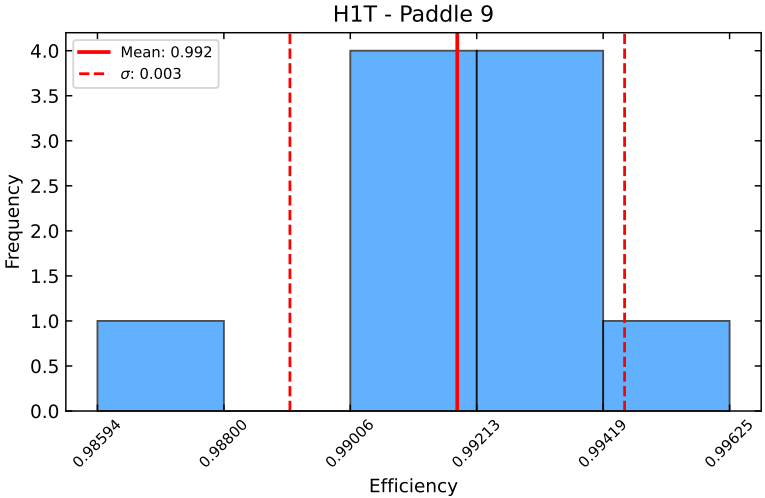
Efficiency Distribution: H1T - Paddle 7



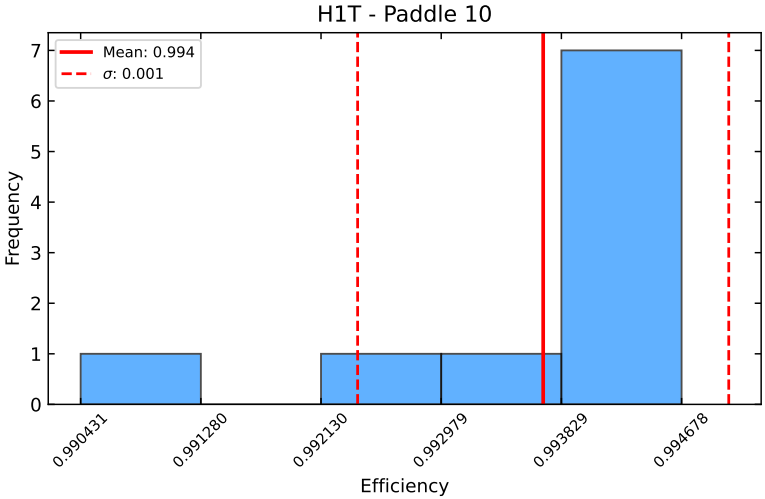
Efficiency Distribution: H1T - Paddle 8



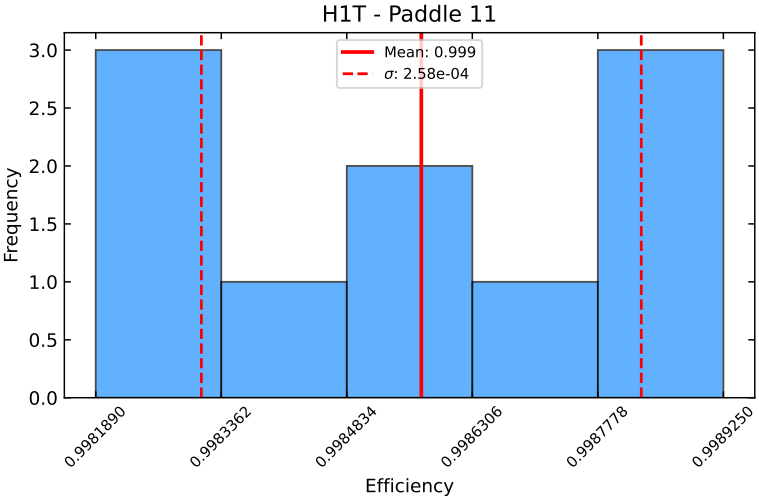
Efficiency Distribution: H1T - Paddle 9



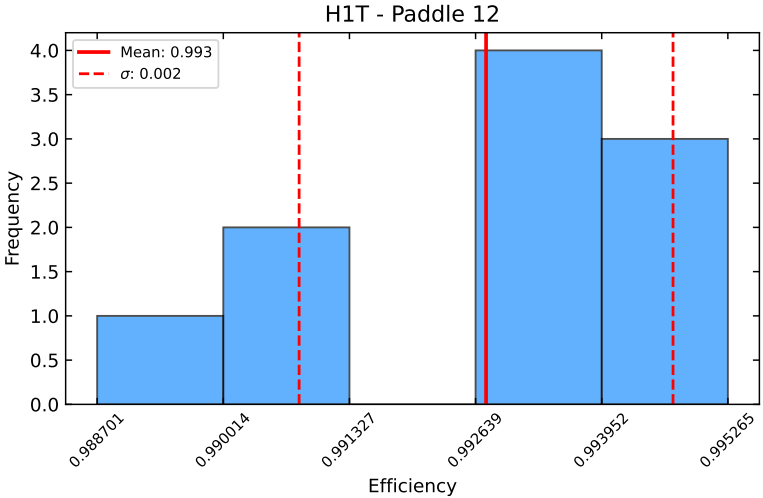
Efficiency Distribution: H1T - Paddle 10



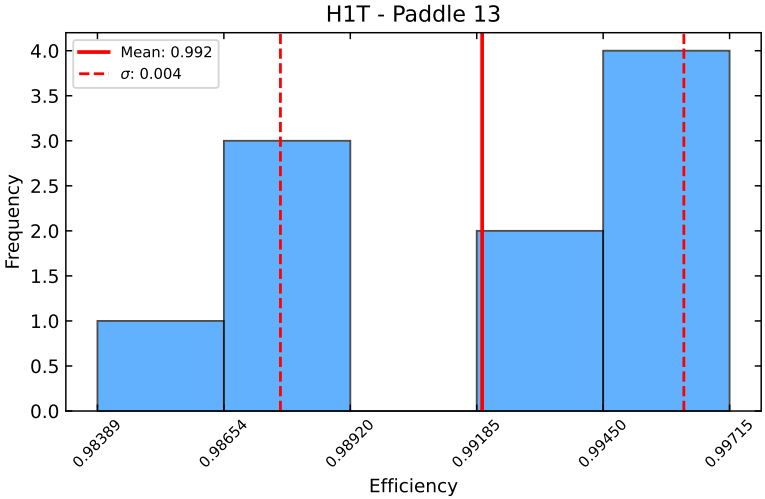
Efficiency Distribution: H1T - Paddle 11



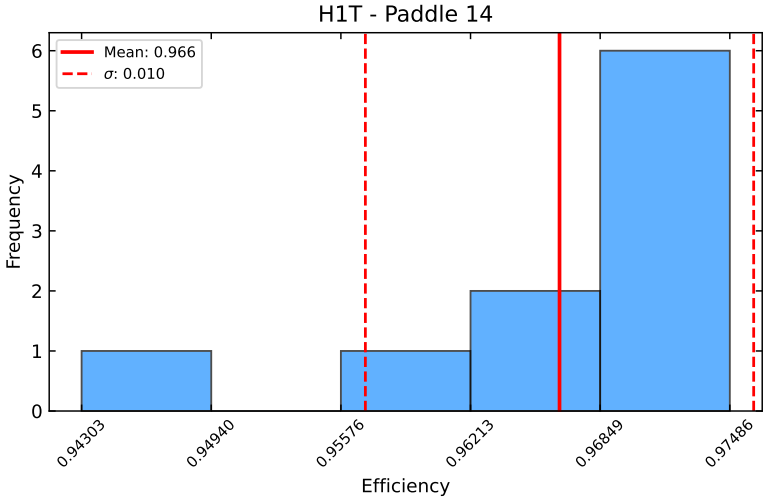
Efficiency Distribution: H1T - Paddle 12



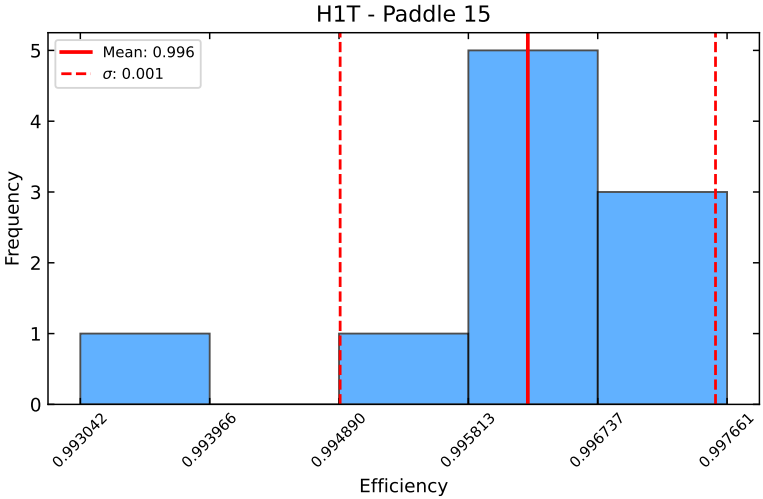
Efficiency Distribution: H1T - Paddle 13



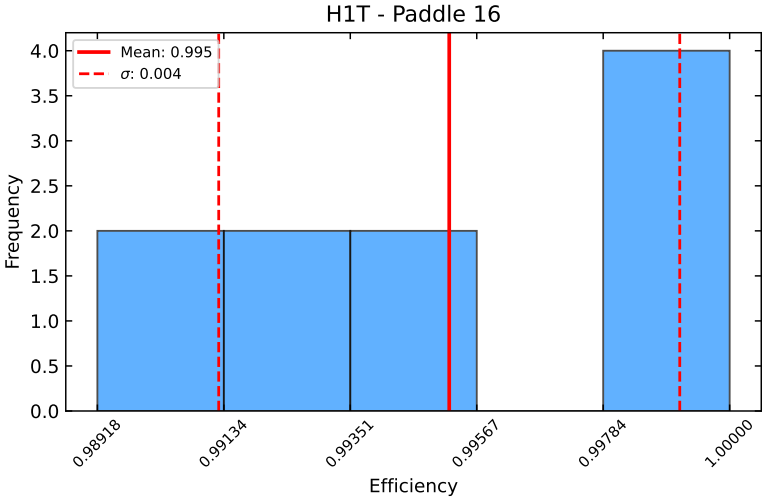
Efficiency Distribution: H1T - Paddle 14



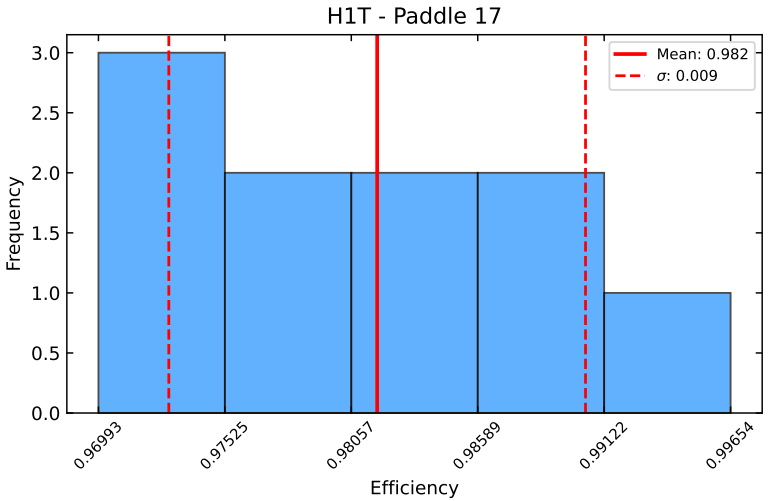
Efficiency Distribution: H1T - Paddle 15



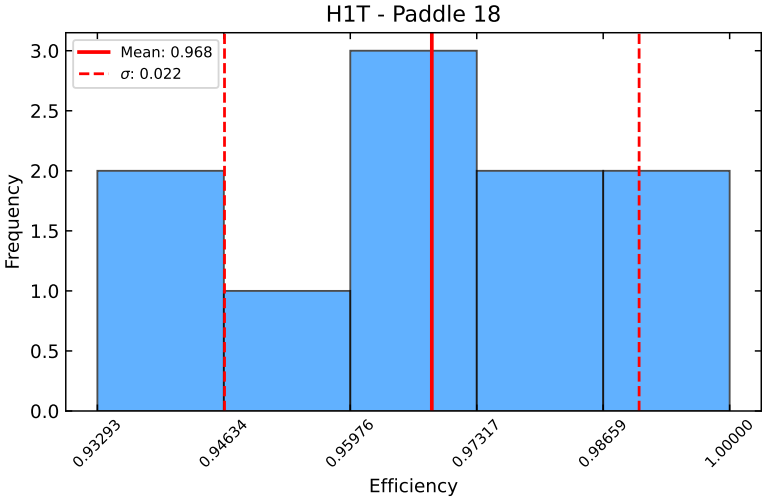
Efficiency Distribution: H1T - Paddle 16



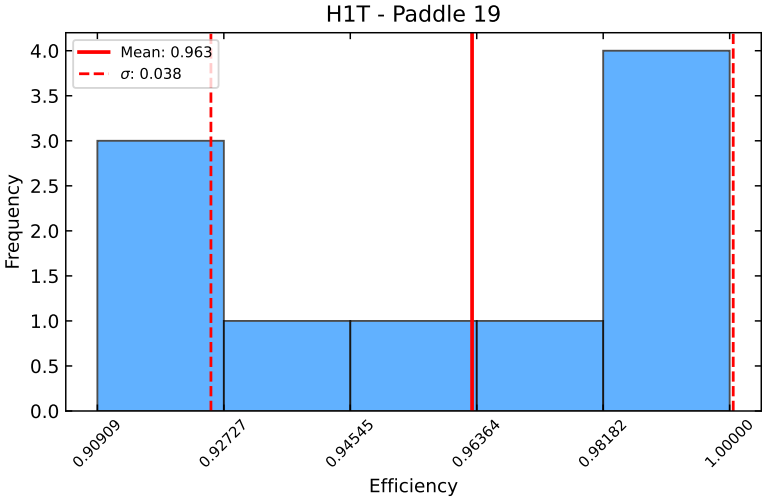
Efficiency Distribution: H1T - Paddle 17



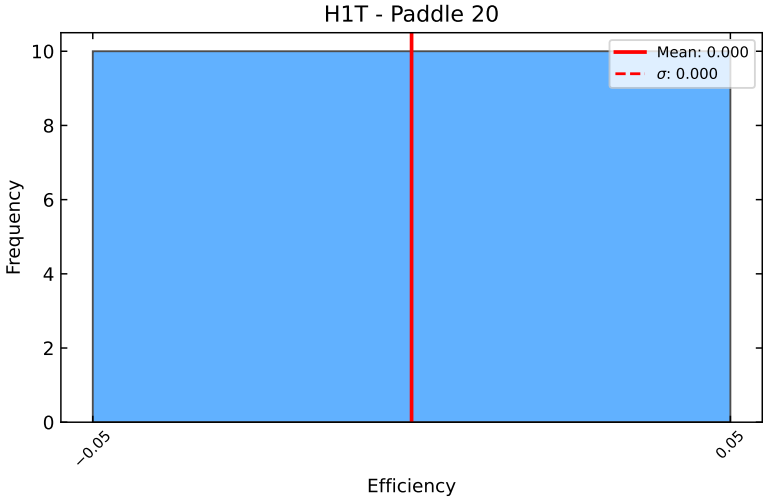
Efficiency Distribution: H1T - Paddle 18



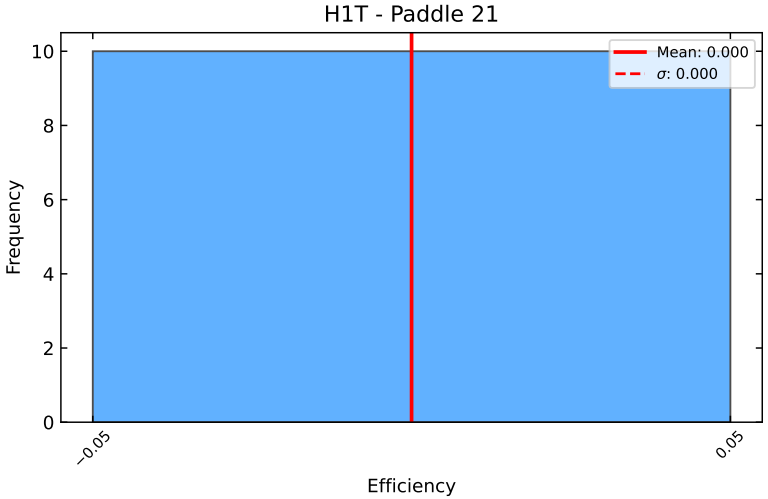
Efficiency Distribution: H1T - Paddle 19



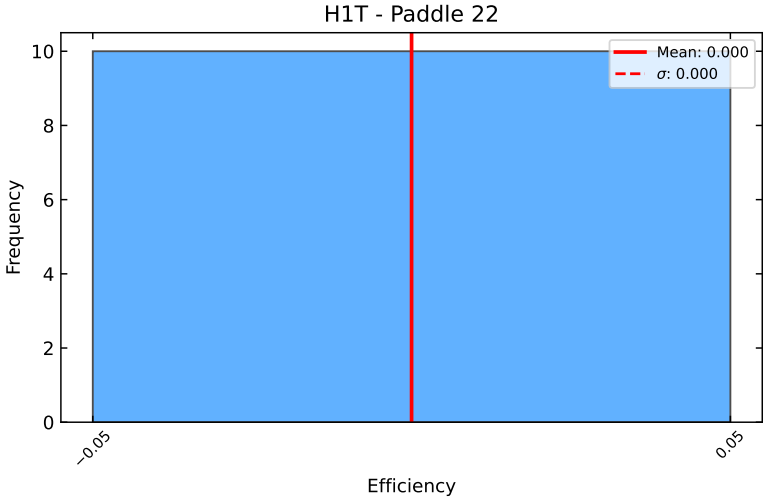
Efficiency Distribution: H1T - Paddle 20



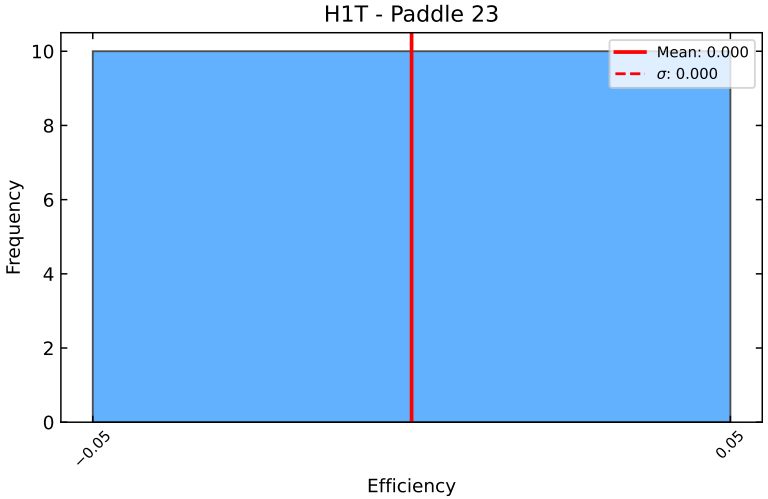
Efficiency Distribution: H1T - Paddle 21



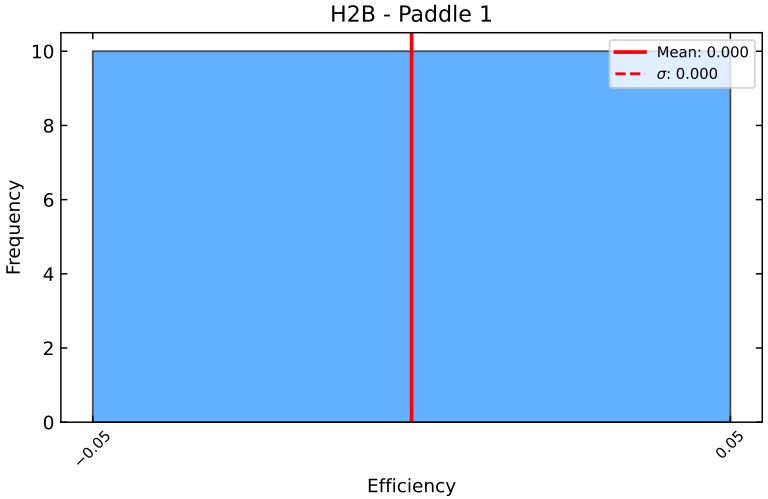
Efficiency Distribution: H1T - Paddle 22



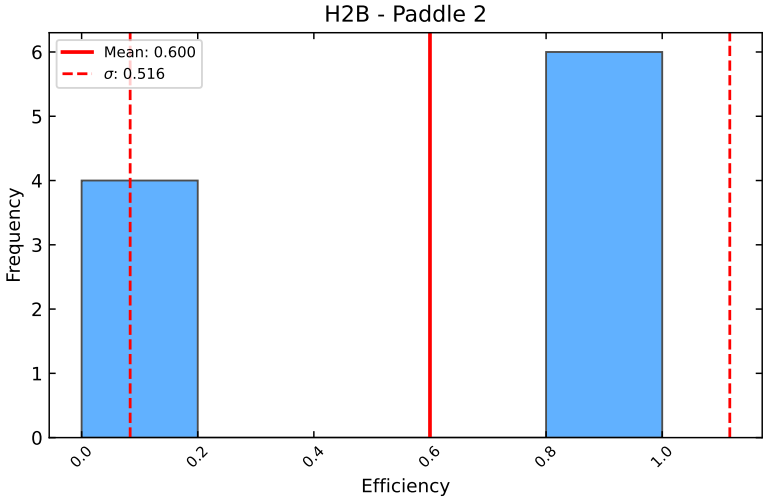
Efficiency Distribution: H1T - Paddle 23



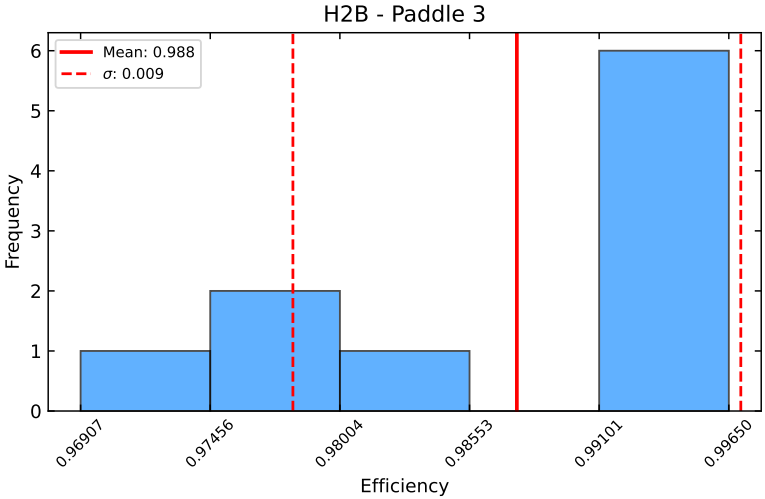
Efficiency Distribution: H2B - Paddle 1



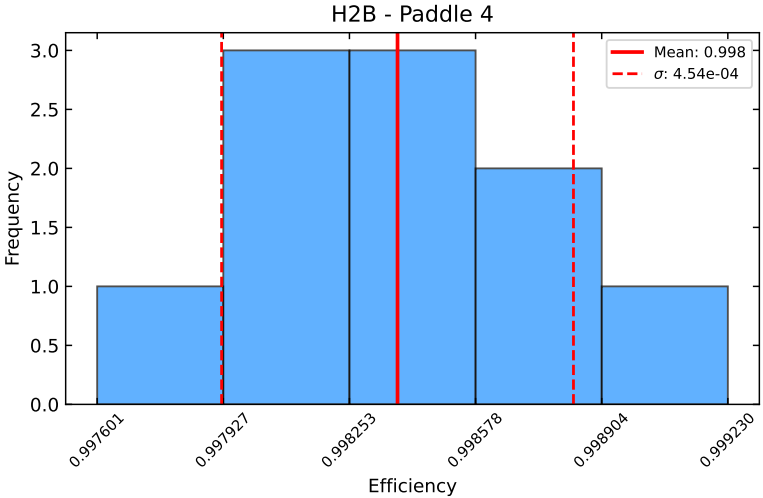
Efficiency Distribution: H2B - Paddle 2



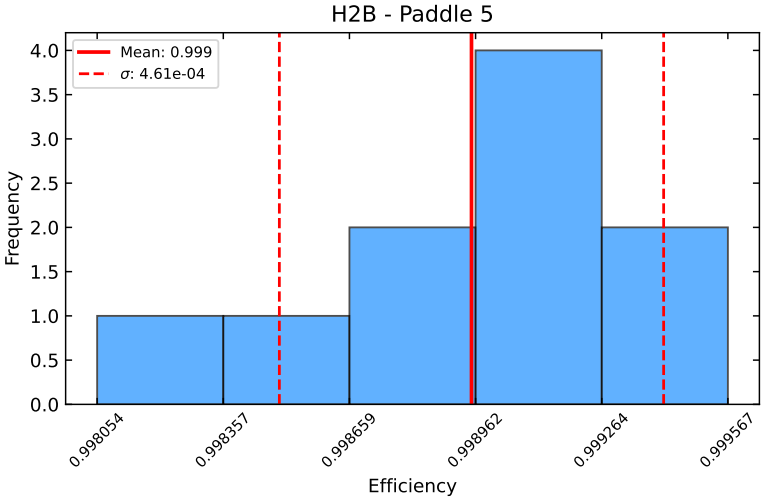
Efficiency Distribution: H2B - Paddle 3



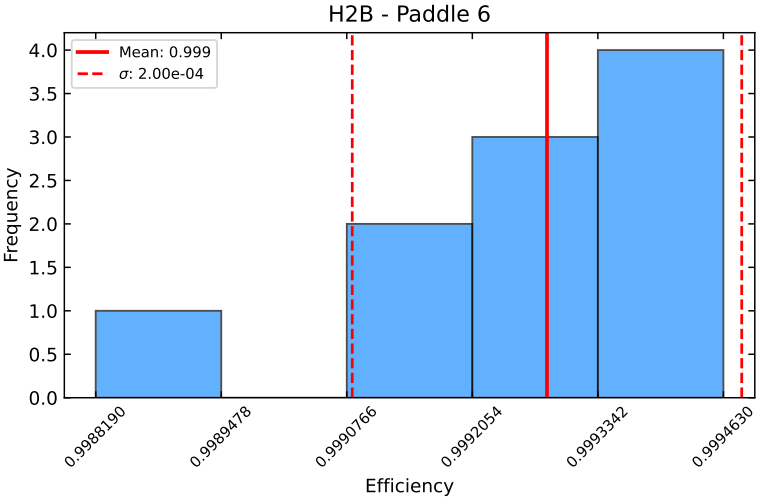
Efficiency Distribution: H2B - Paddle 4



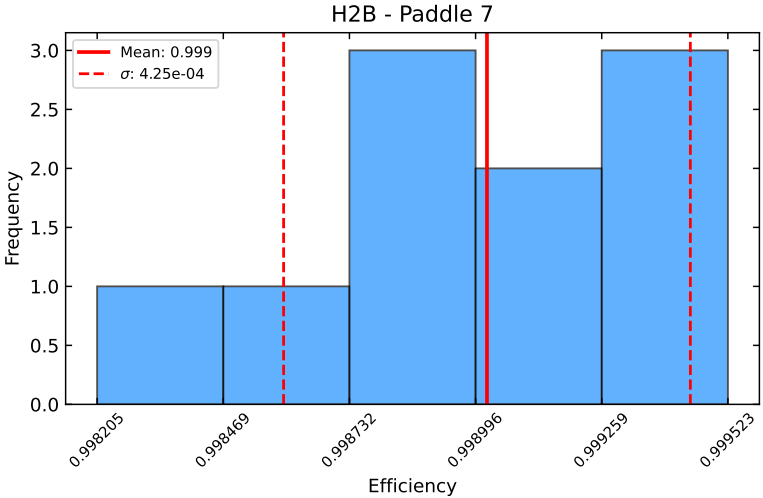
Efficiency Distribution: H2B - Paddle 5



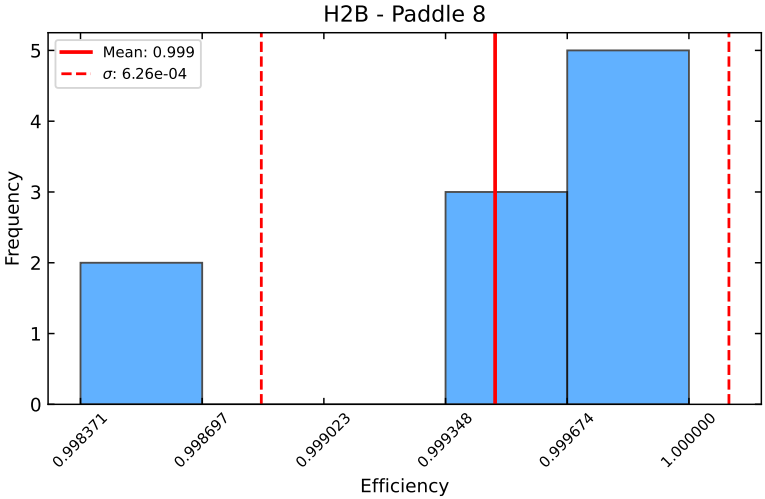
Efficiency Distribution: H2B - Paddle 6



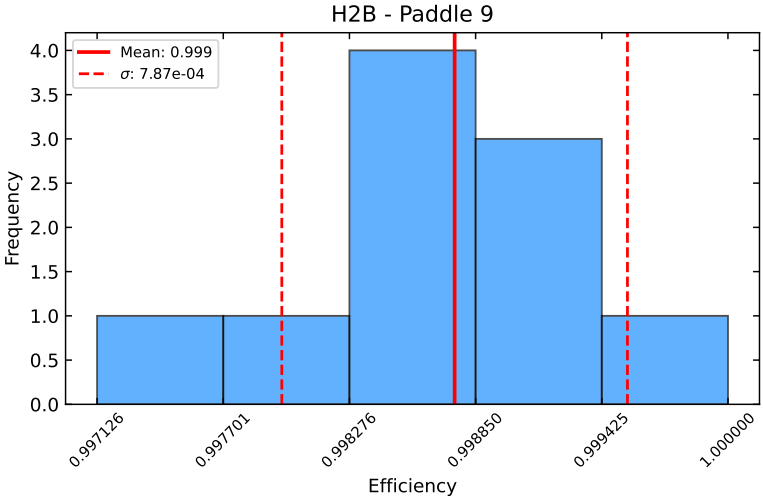
Efficiency Distribution: H2B - Paddle 7



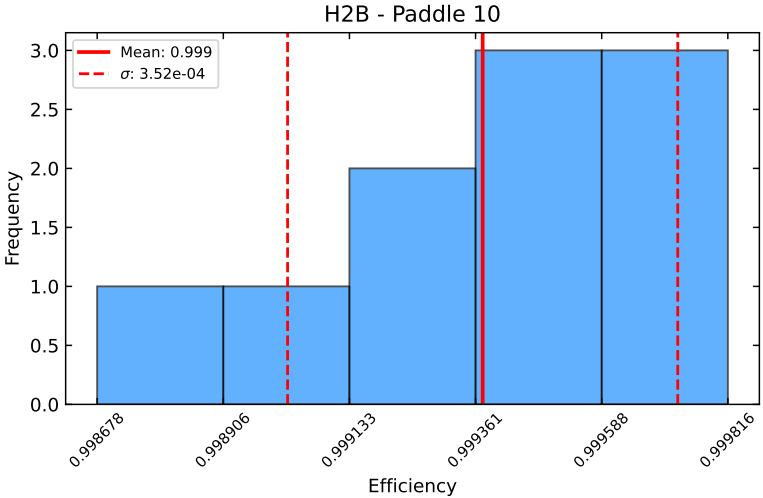
Efficiency Distribution: H2B - Paddle 8



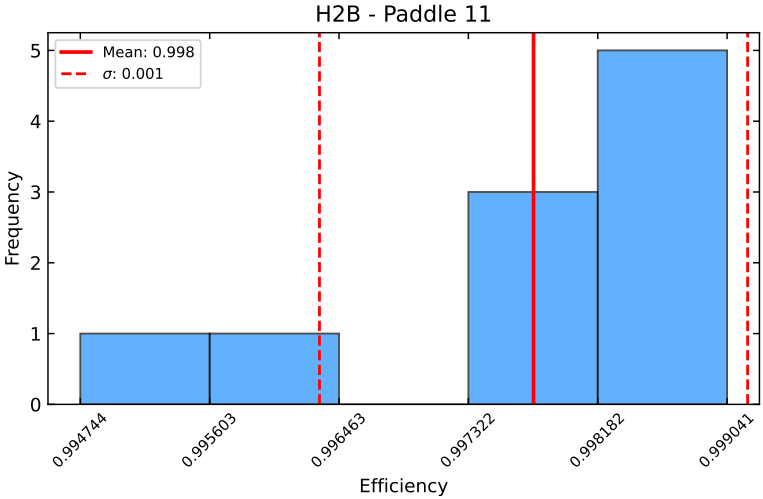
Efficiency Distribution: H2B - Paddle 9



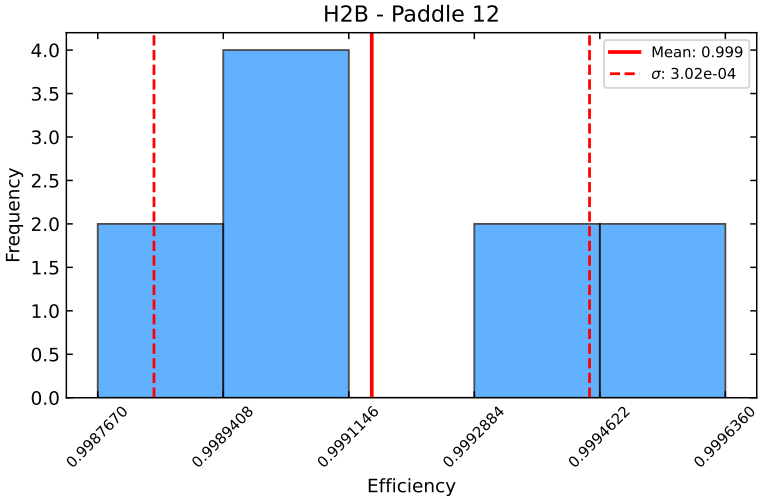
Efficiency Distribution: H2B - Paddle 10



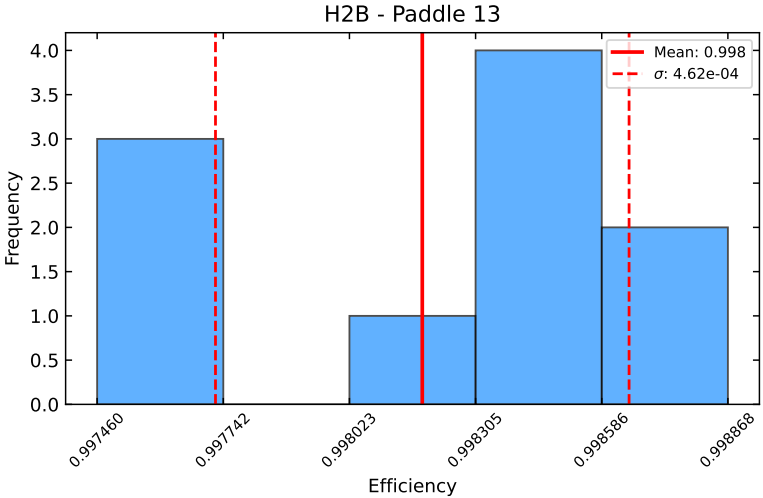
Efficiency Distribution: H2B - Paddle 11



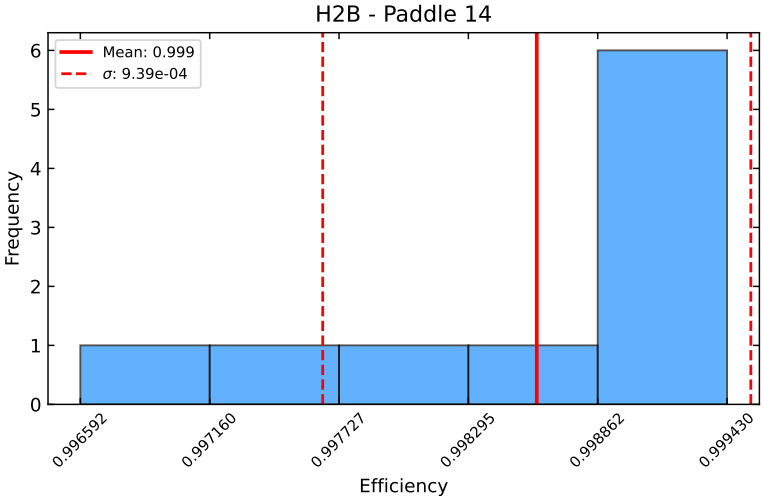
Efficiency Distribution: H2B - Paddle 12



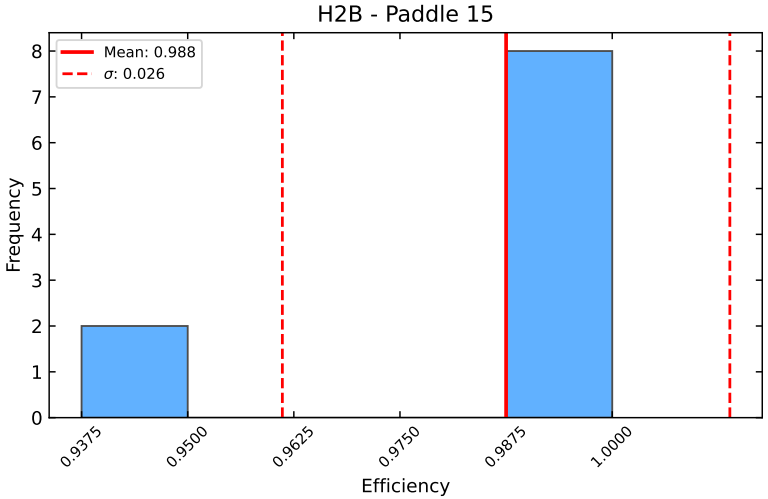
Efficiency Distribution: H2B - Paddle 13



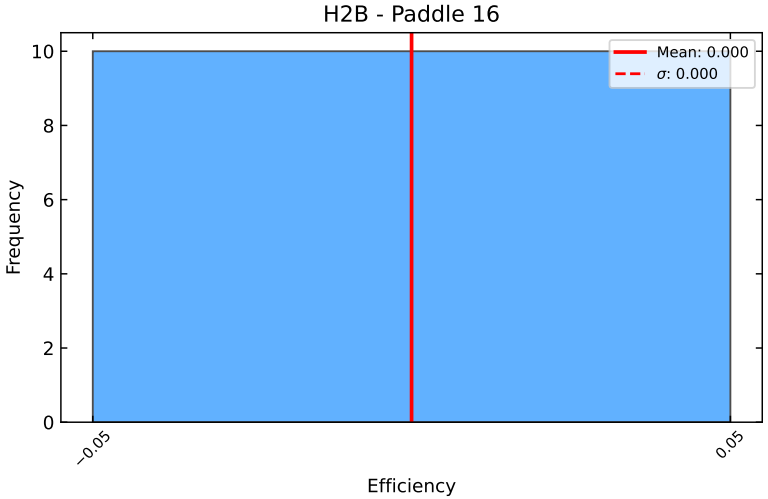
Efficiency Distribution: H2B - Paddle 14



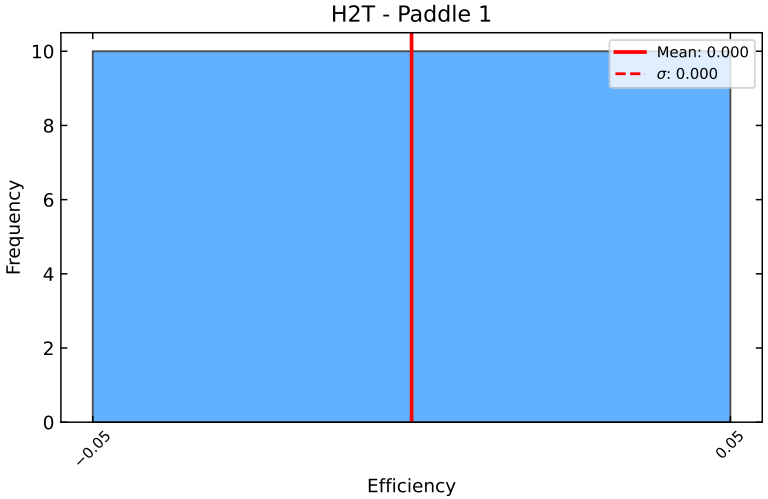
Efficiency Distribution: H2B - Paddle 15



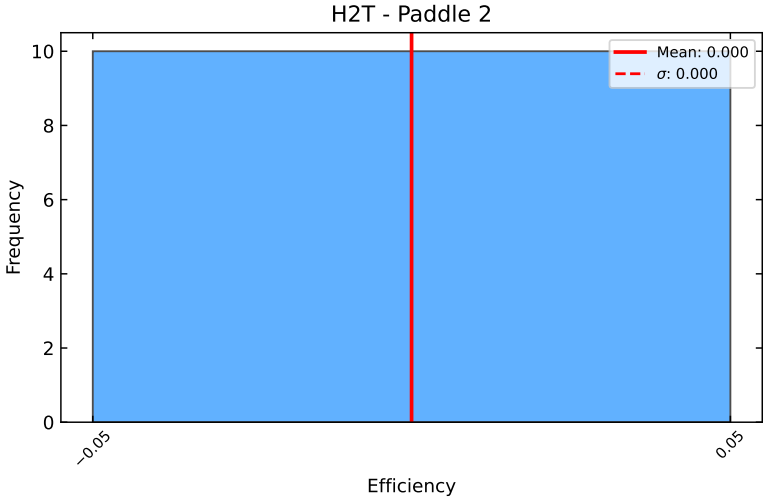
Efficiency Distribution: H2B - Paddle 16



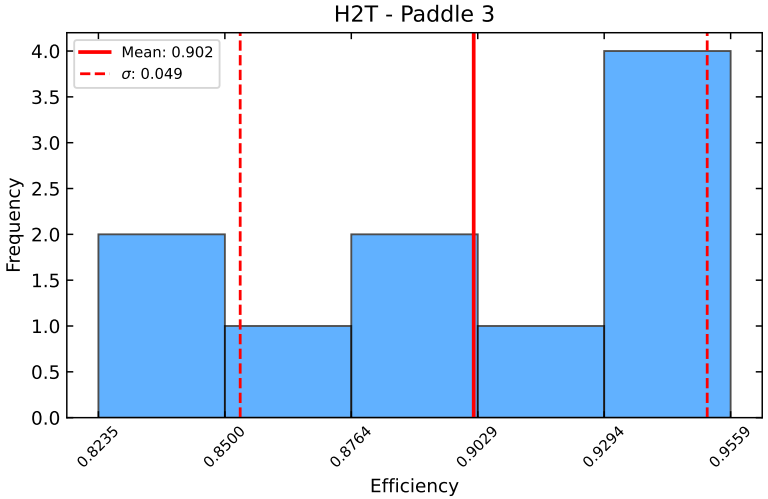
Efficiency Distribution: H2T - Paddle 1



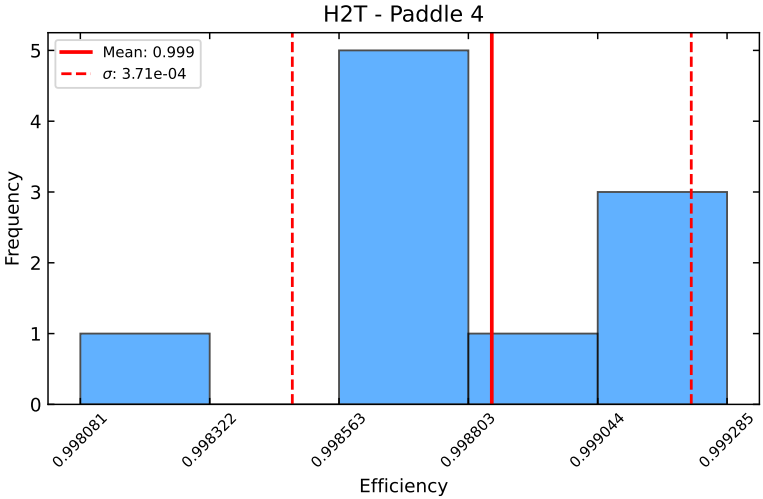
Efficiency Distribution: H2T - Paddle 2



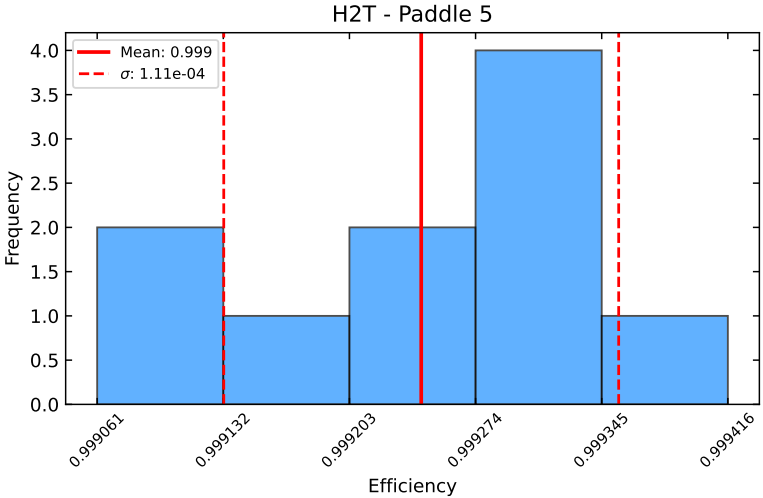
Efficiency Distribution: H2T - Paddle 3



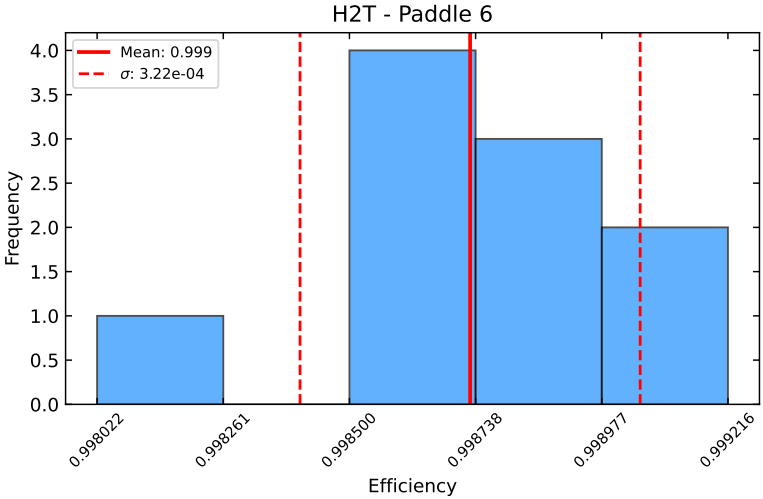
Efficiency Distribution: H2T - Paddle 4



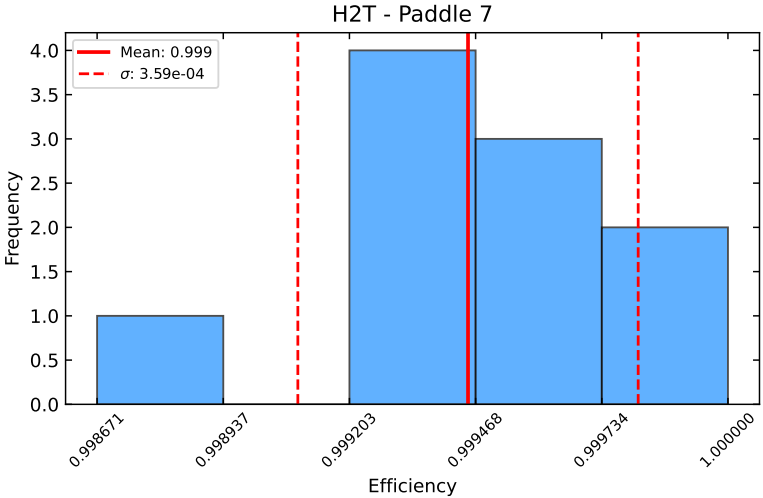
Efficiency Distribution: H2T - Paddle 5



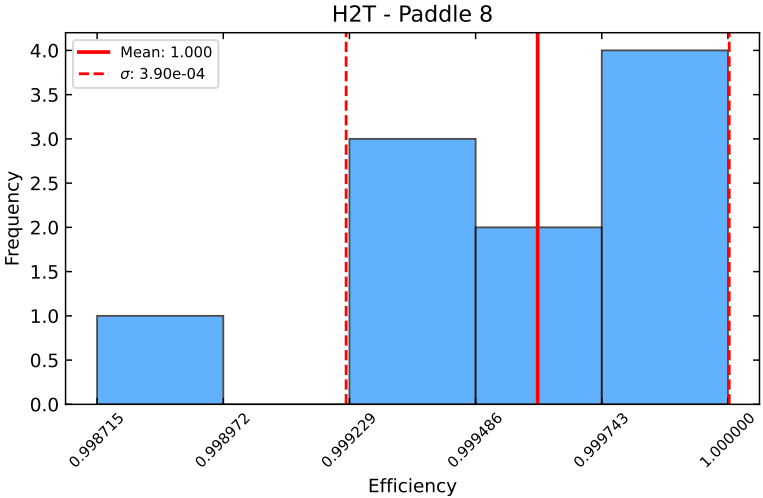
Efficiency Distribution: H2T - Paddle 6



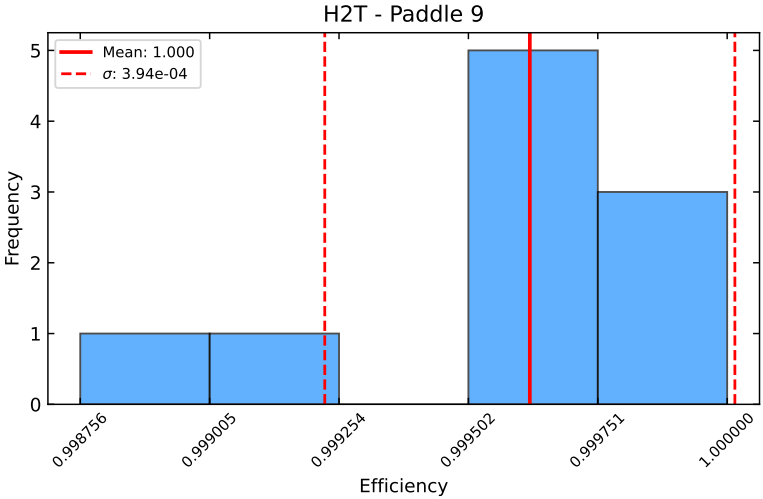
Efficiency Distribution: H2T - Paddle 7



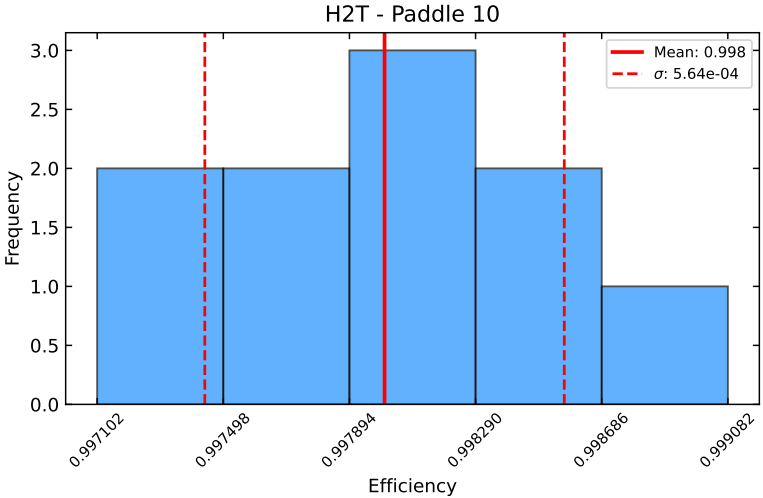
Efficiency Distribution: H2T - Paddle 8



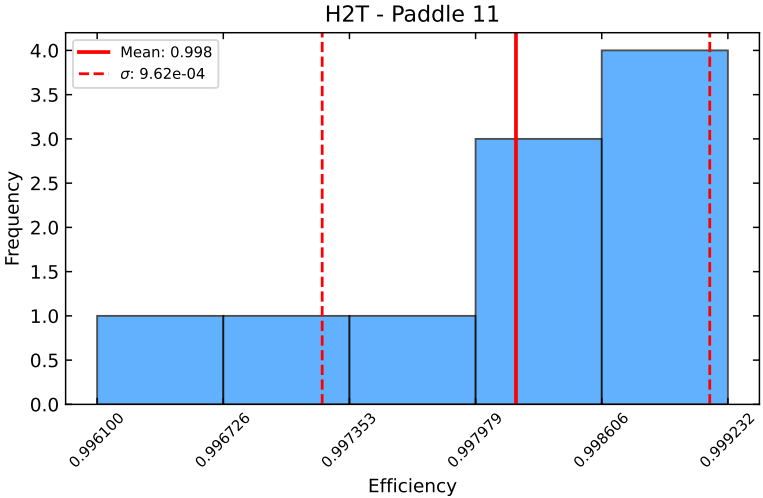
Efficiency Distribution: H2T - Paddle 9



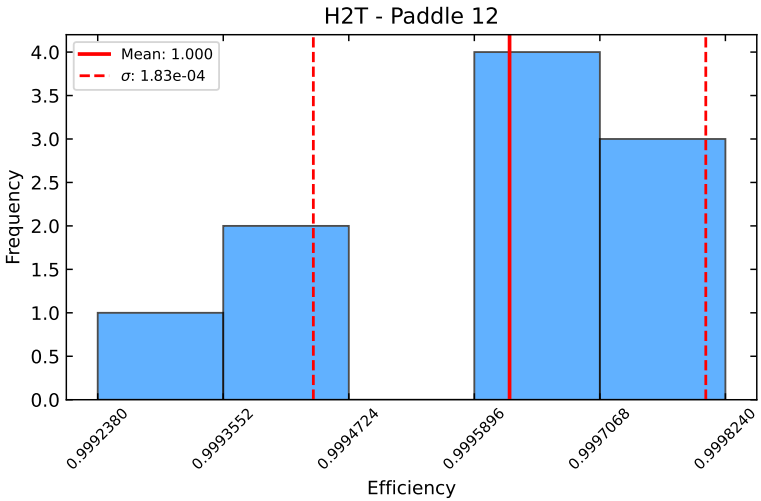
Efficiency Distribution: H2T - Paddle 10



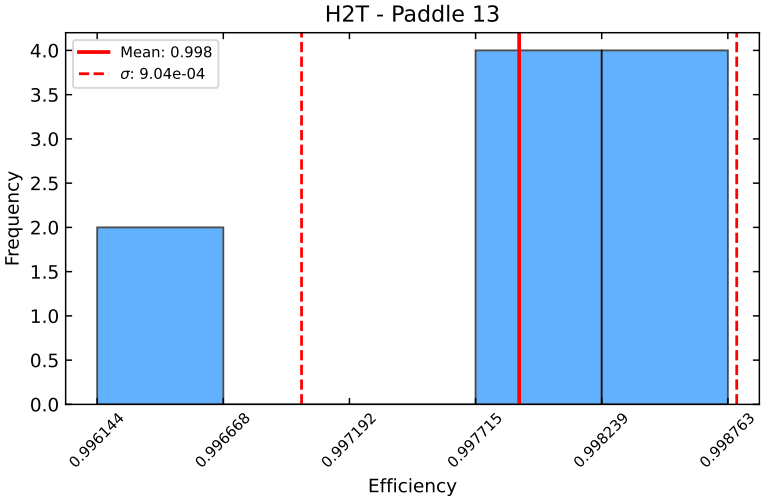
Efficiency Distribution: H2T - Paddle 11



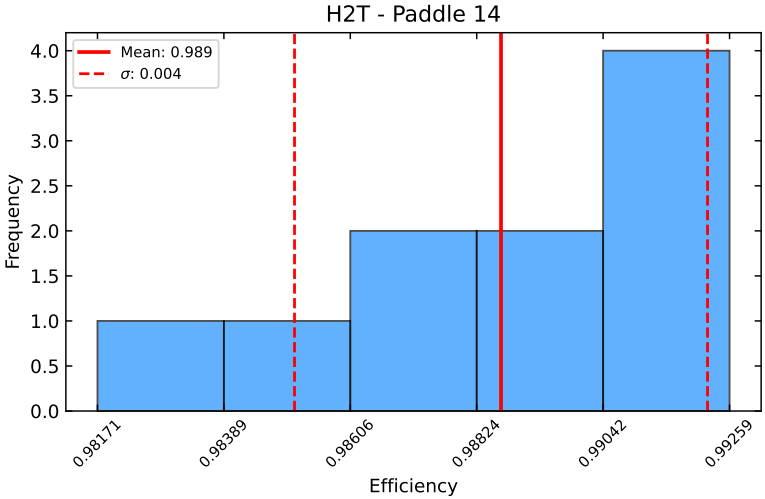
Efficiency Distribution: H2T - Paddle 12



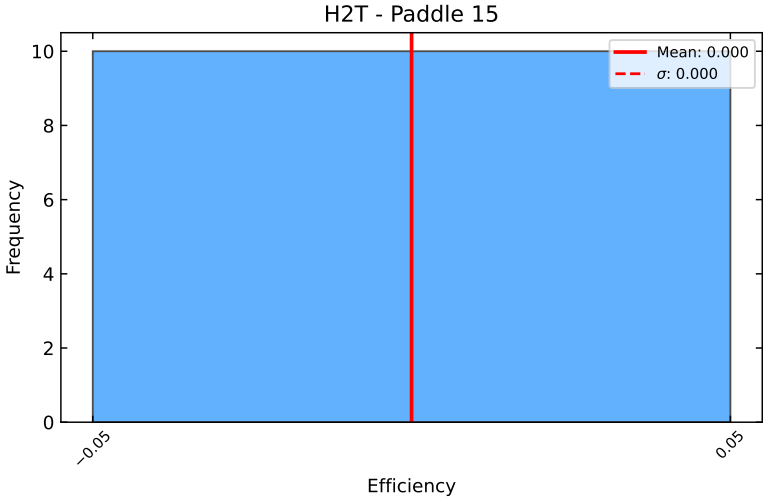
Efficiency Distribution: H2T - Paddle 13



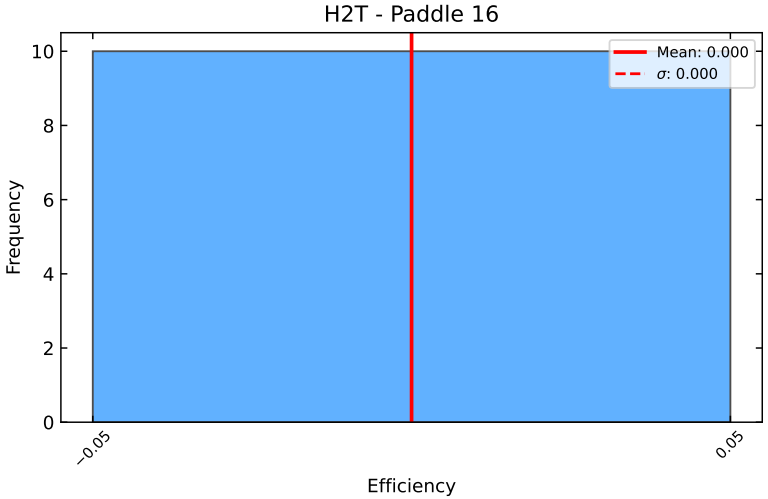
Efficiency Distribution: H2T - Paddle 14



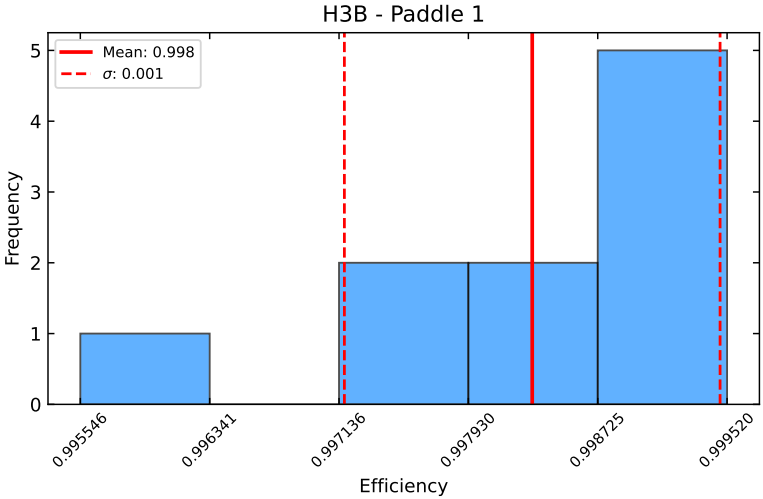
Efficiency Distribution: H2T - Paddle 15



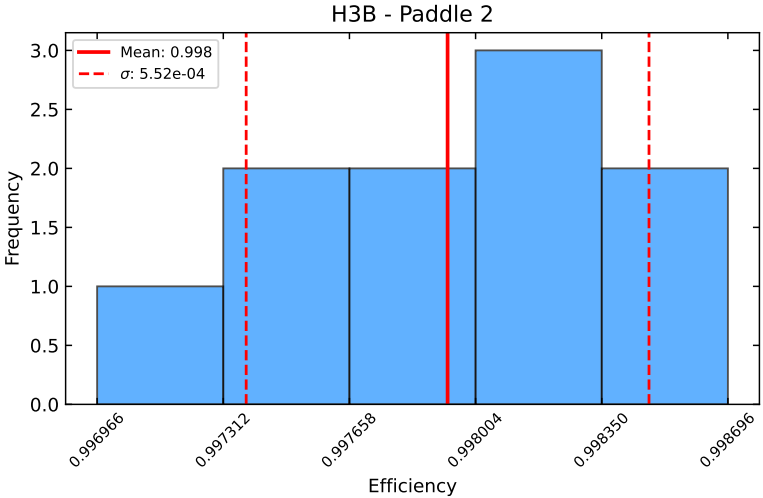
Efficiency Distribution: H2T - Paddle 16



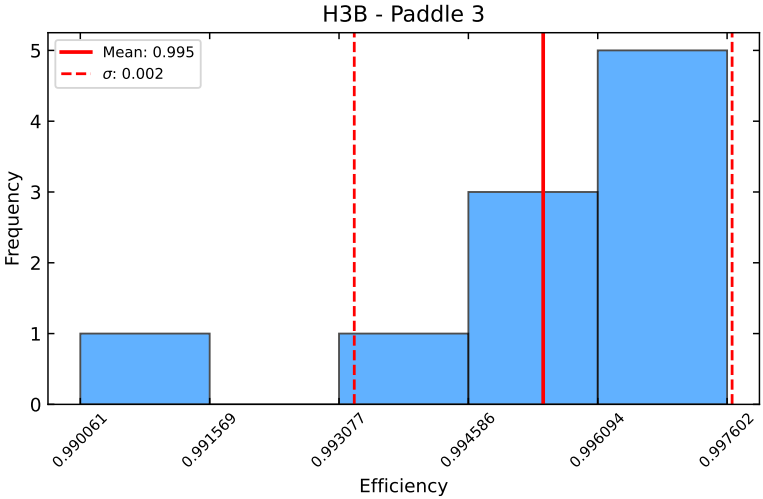
Efficiency Distribution: H3B - Paddle 1



Efficiency Distribution: H3B - Paddle 2



Efficiency Distribution: H3B - Paddle 3



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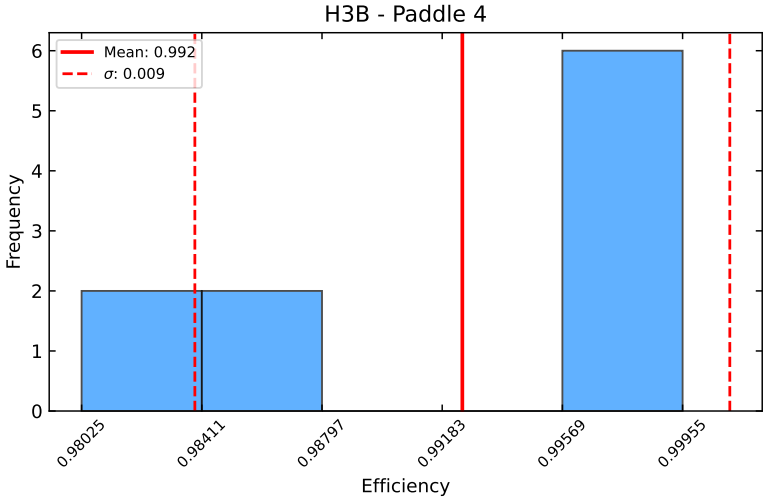
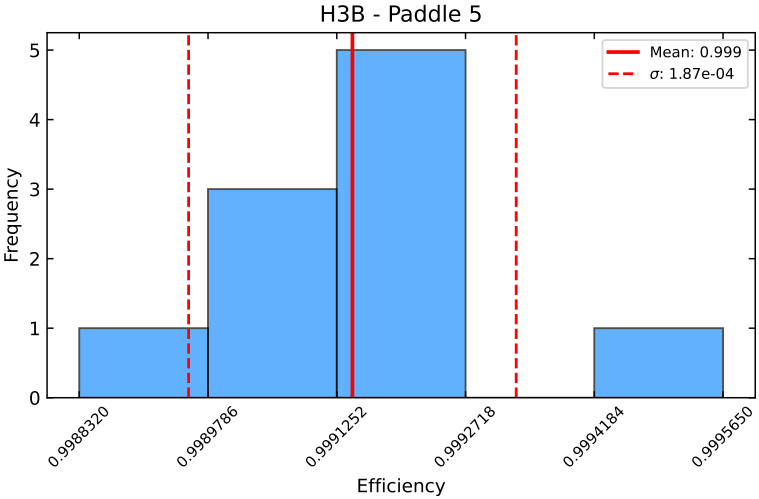
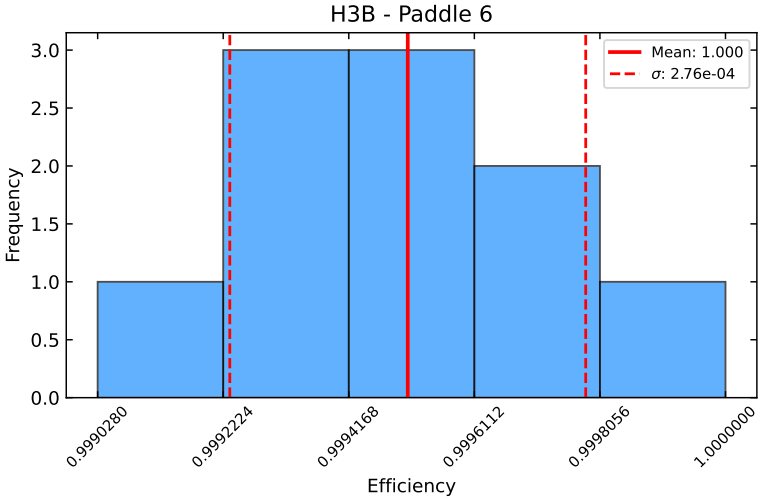


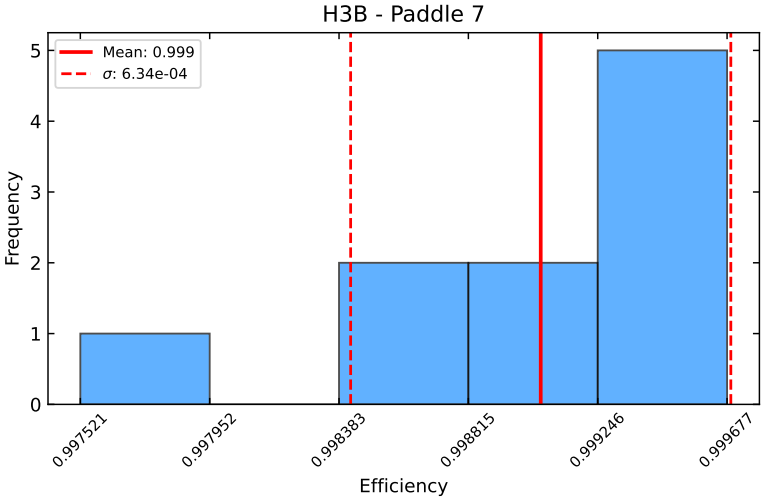
Efficiency Distribution: H3B - Paddle 5



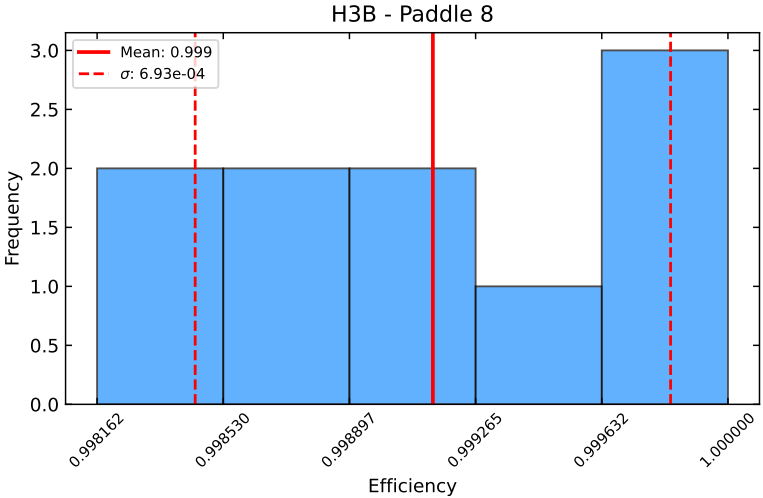
Efficiency Distribution: H3B - Paddle 6



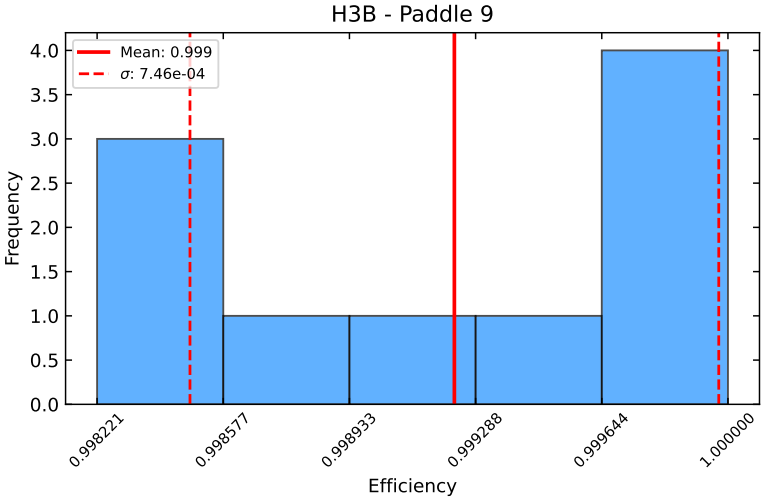
Efficiency Distribution: H3B - Paddle 7



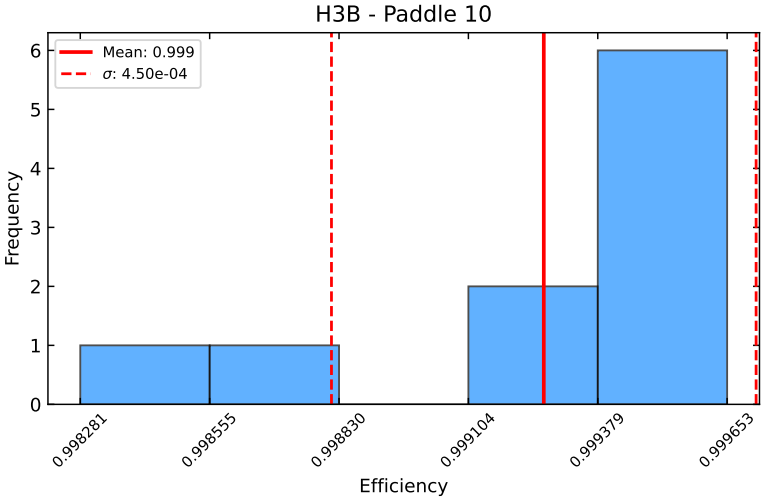
Efficiency Distribution: H3B - Paddle 8



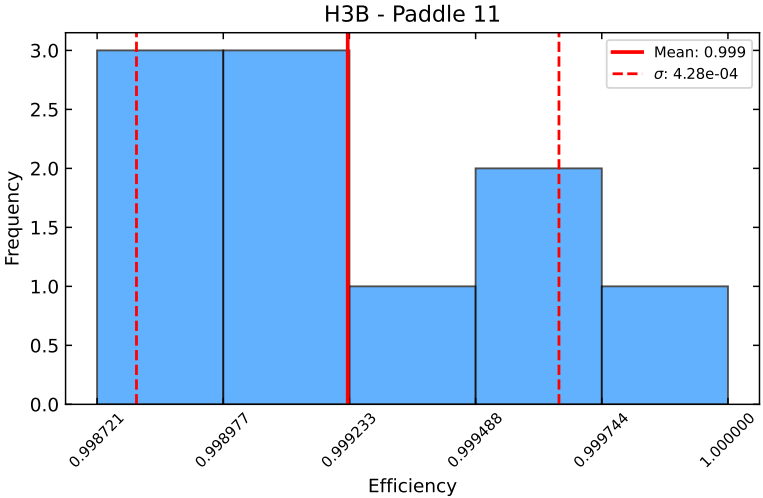
Efficiency Distribution: H3B - Paddle 9



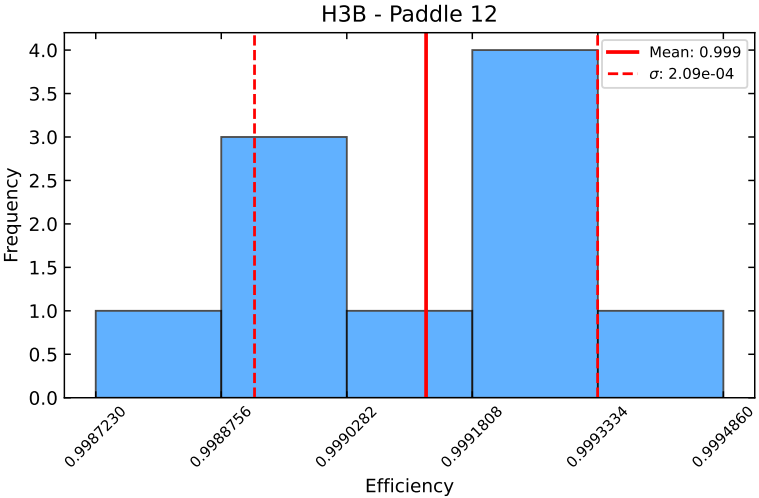
Efficiency Distribution: H3B - Paddle 10



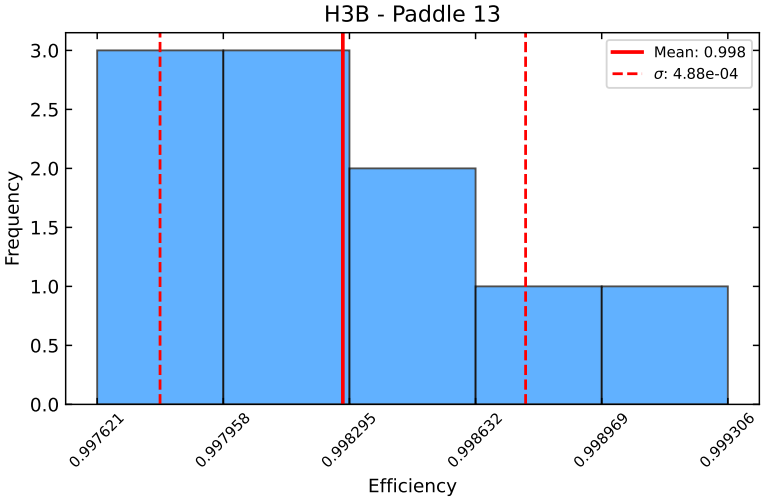
Efficiency Distribution: H3B - Paddle 11



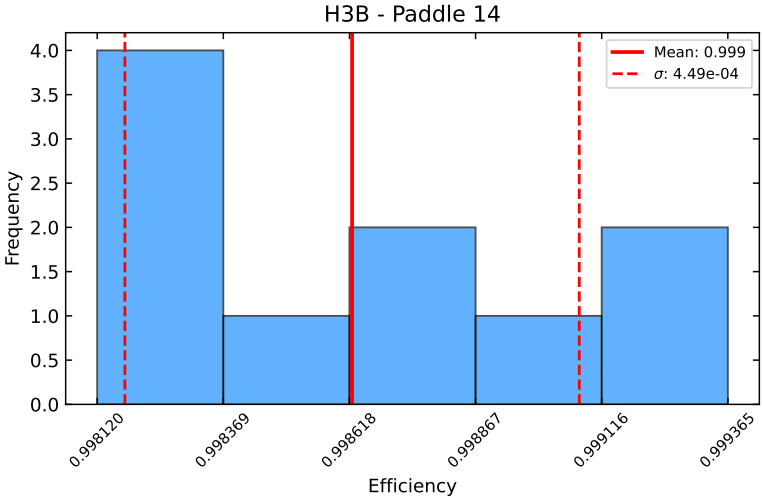
Efficiency Distribution: H3B - Paddle 12



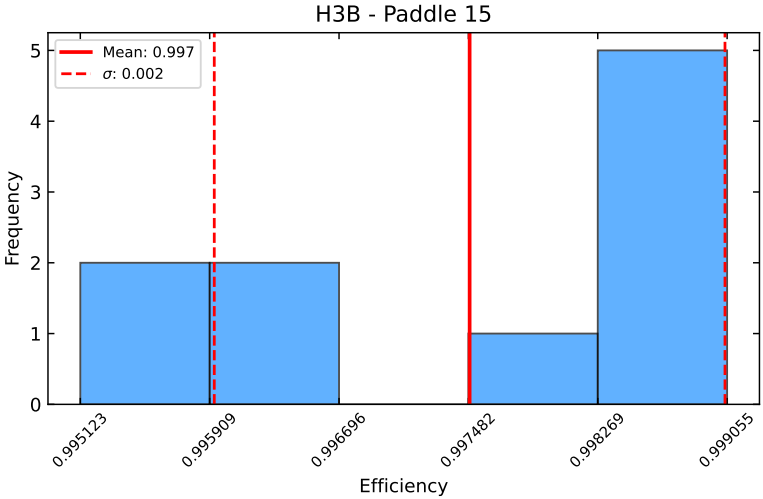
Efficiency Distribution: H3B - Paddle 13



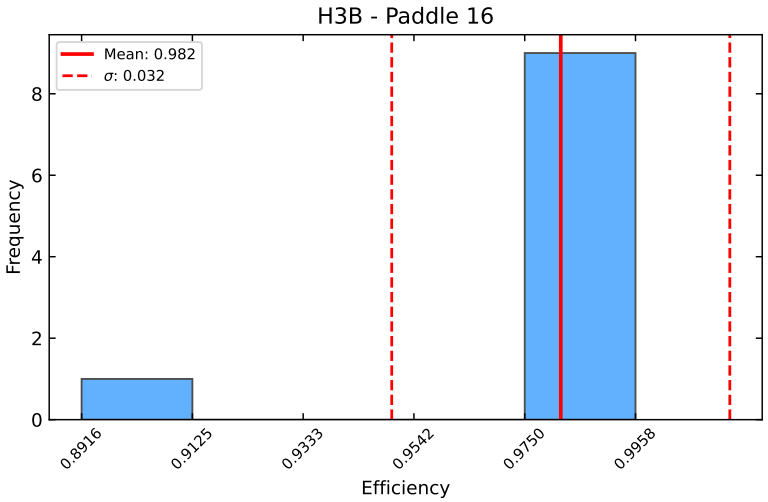
Efficiency Distribution: H3B - Paddle 14



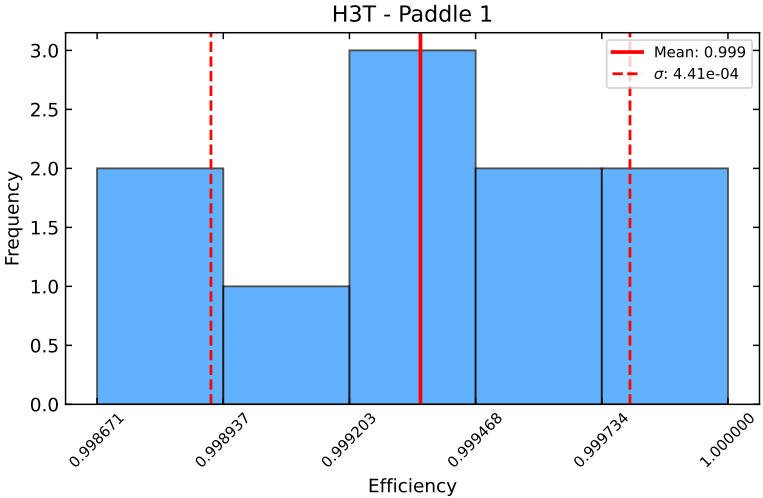
Efficiency Distribution: H3B - Paddle 15



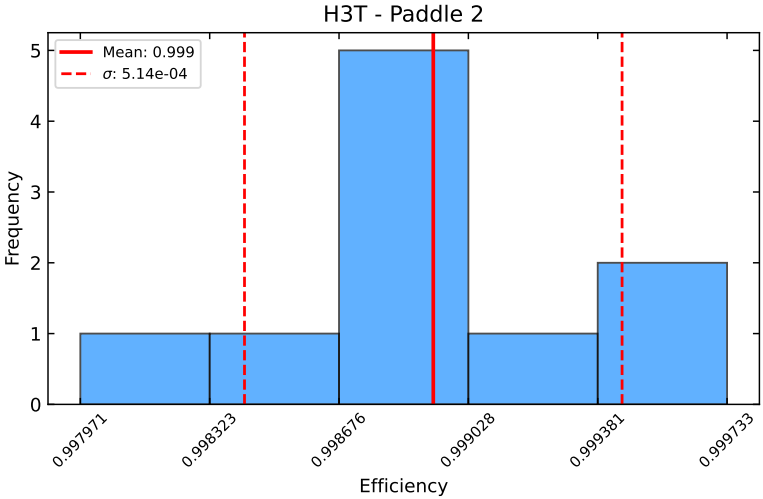
Efficiency Distribution: H3B - Paddle 16



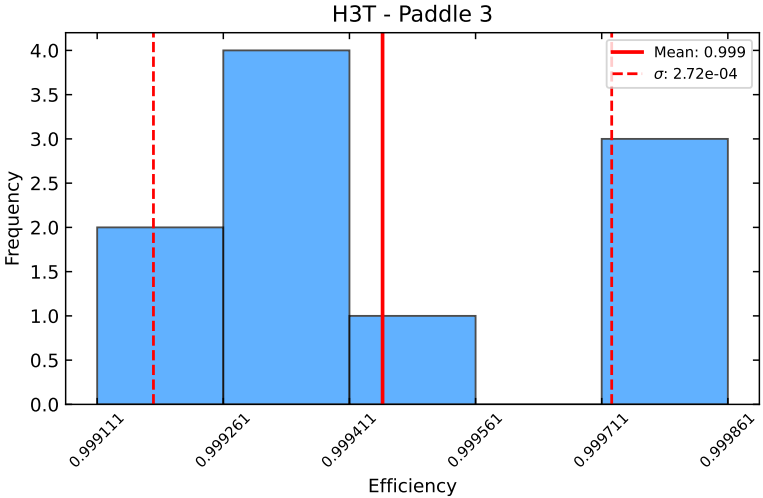
Efficiency Distribution: H3T - Paddle 1



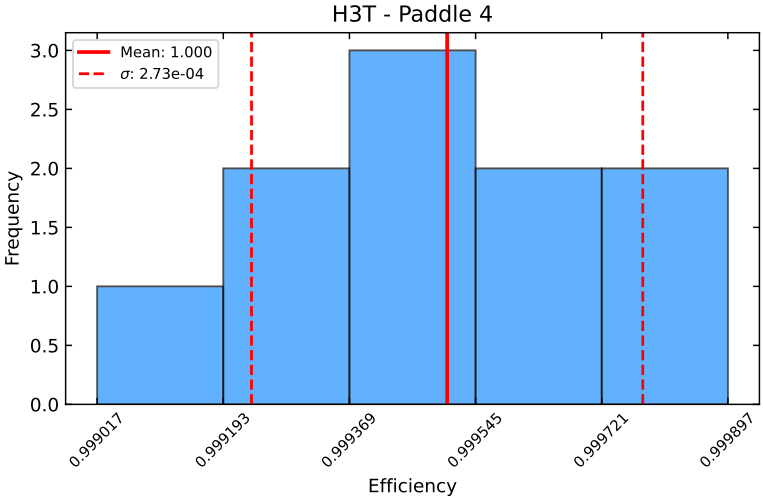
Efficiency Distribution: H3T - Paddle 2



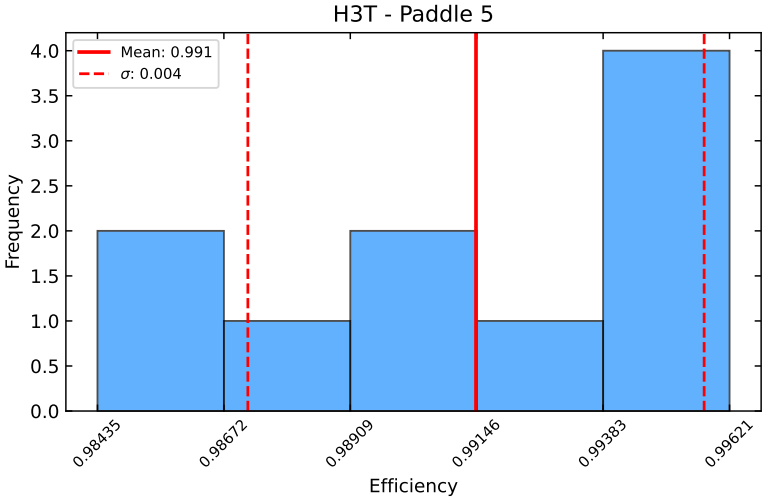
Efficiency Distribution: H3T - Paddle 3



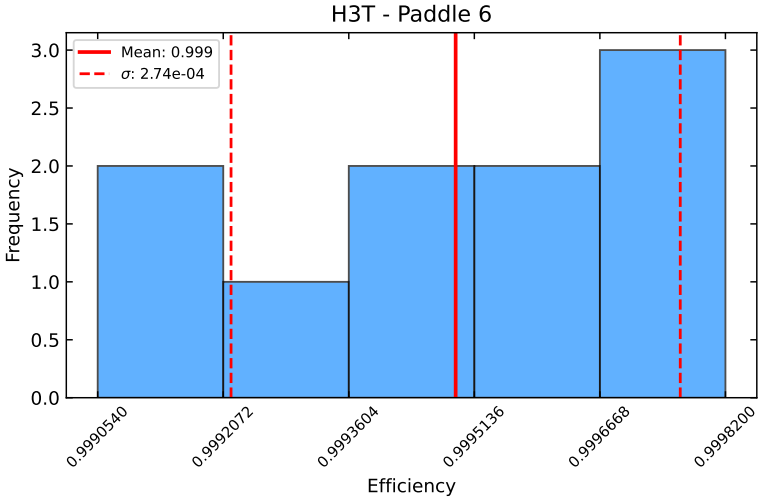
Efficiency Distribution: H3T - Paddle 4



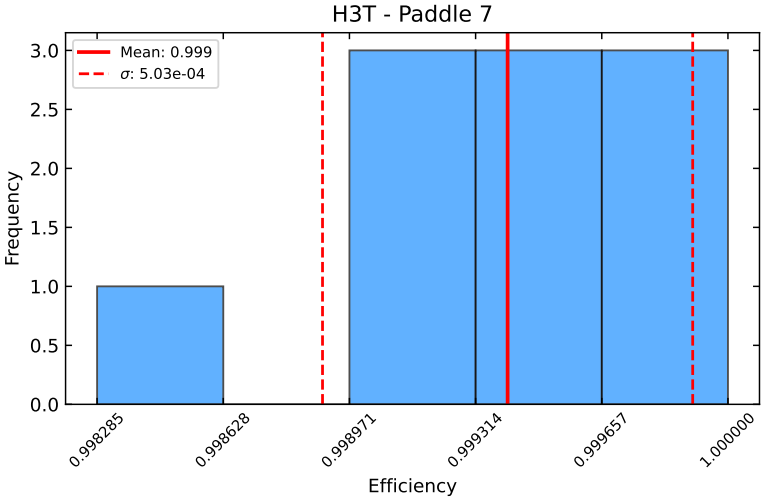
Efficiency Distribution: H3T - Paddle 5



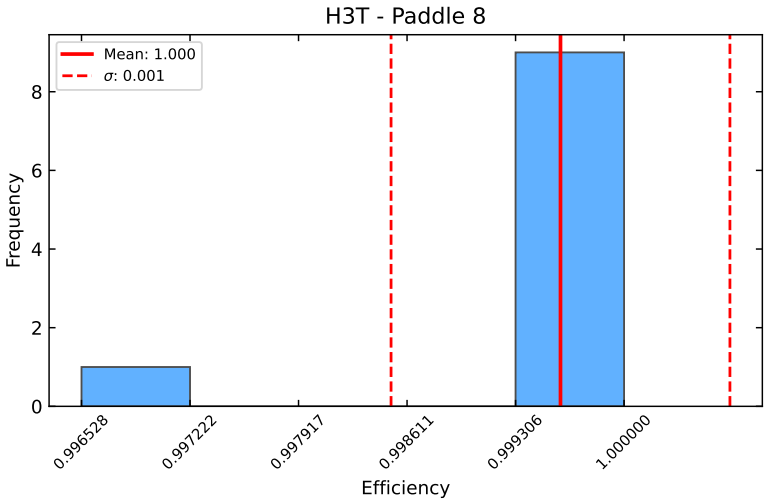
Efficiency Distribution: H3T - Paddle 6



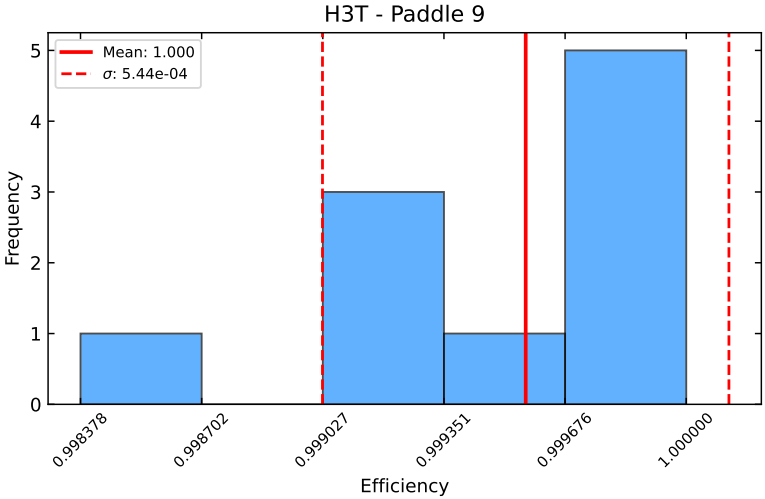
Efficiency Distribution: H3T - Paddle 7



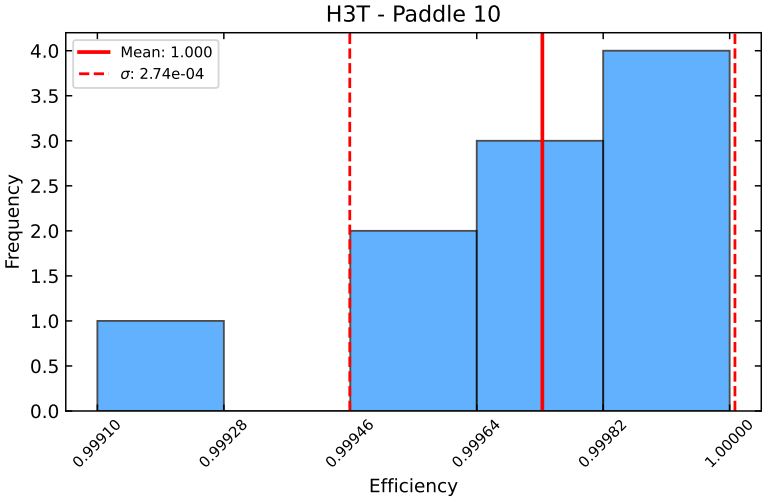
Efficiency Distribution: H3T - Paddle 8



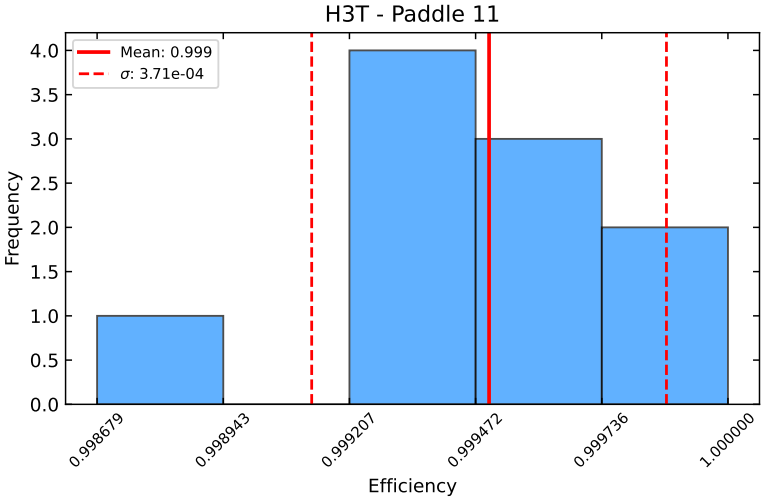
Efficiency Distribution: H3T - Paddle 9



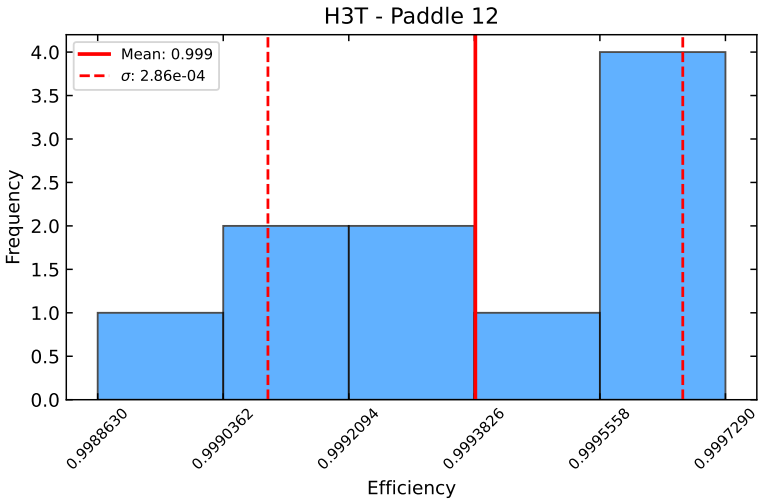
Efficiency Distribution: H3T - Paddle 10



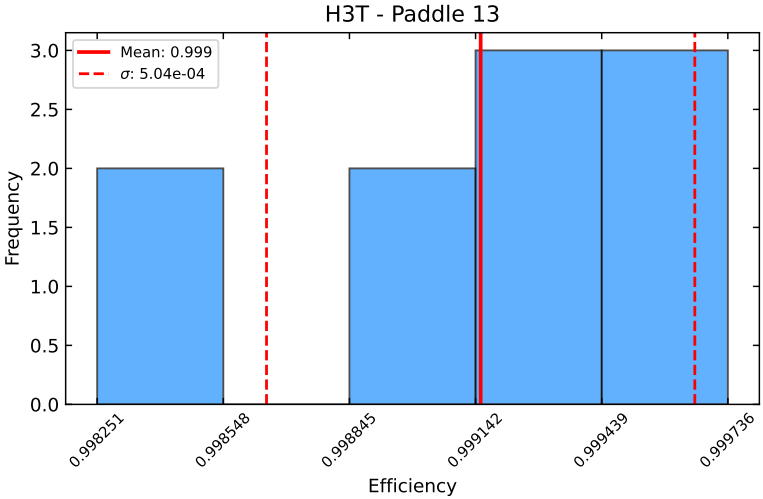
Efficiency Distribution: H3T - Paddle 11



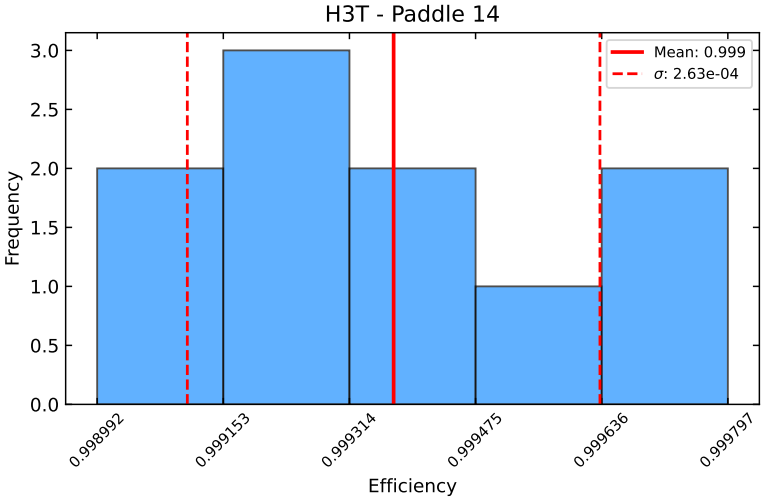
Efficiency Distribution: H3T - Paddle 12



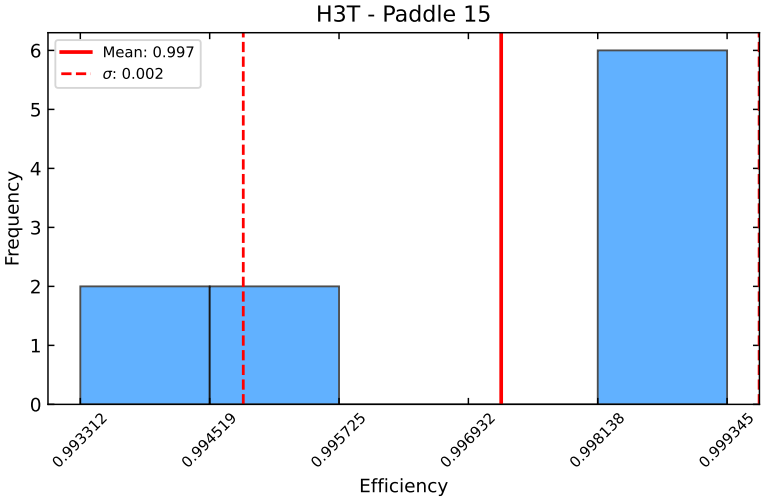
Efficiency Distribution: H3T - Paddle 13



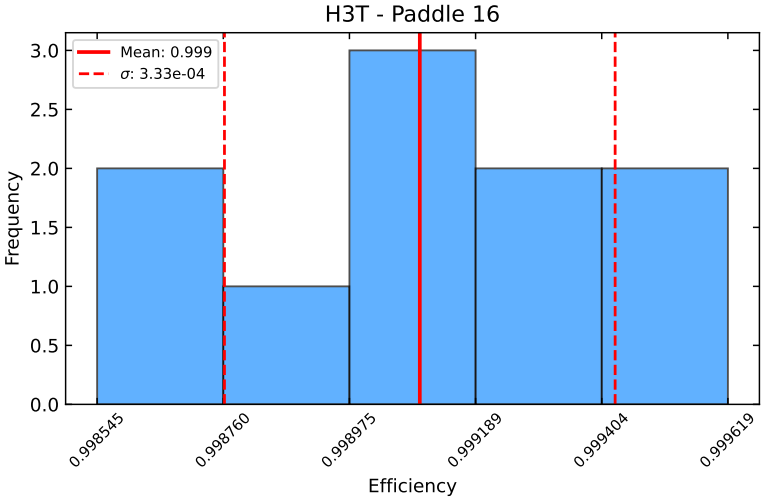
Efficiency Distribution: H3T - Paddle 14



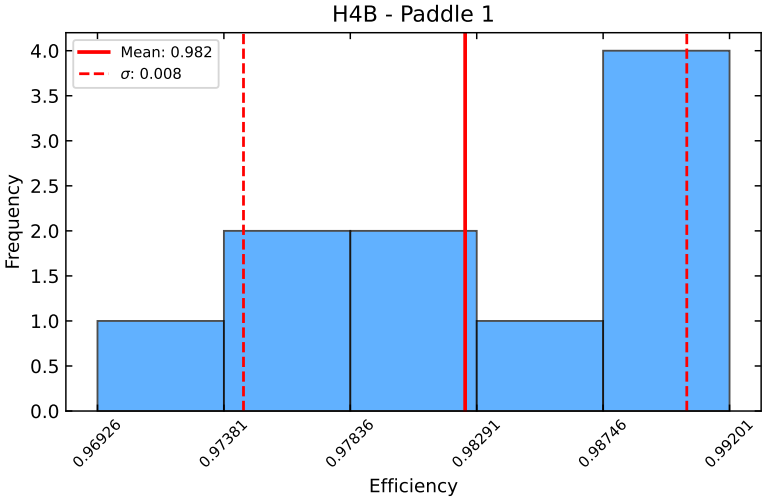
Efficiency Distribution: H3T - Paddle 15



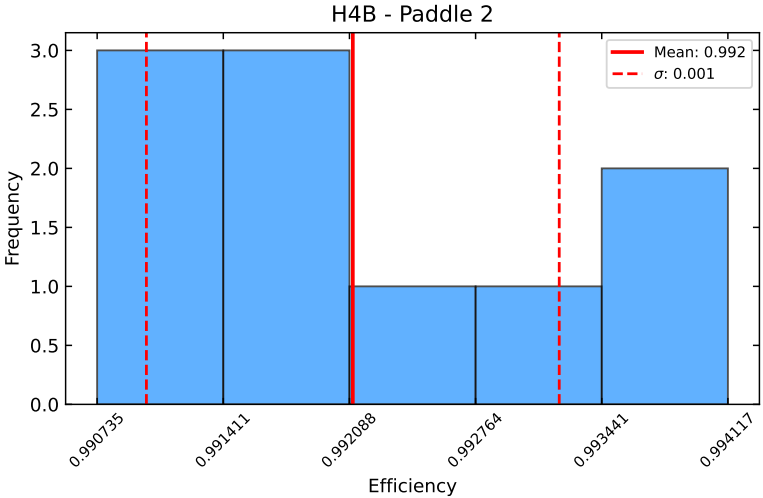
Efficiency Distribution: H3T - Paddle 16



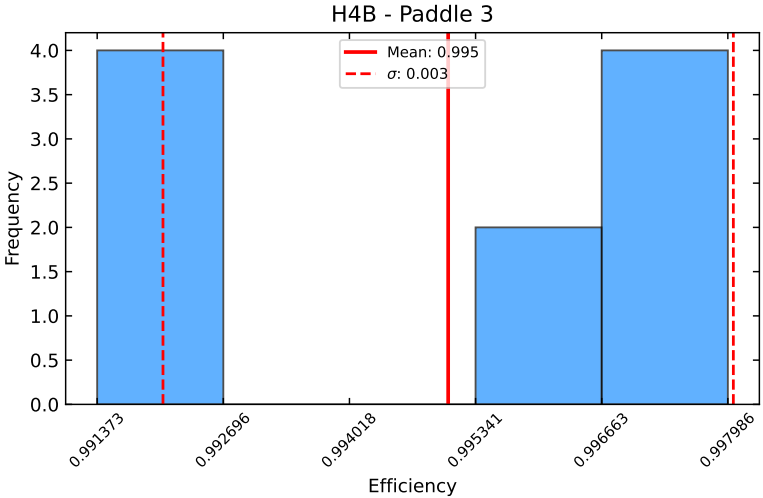
Efficiency Distribution: H4B - Paddle 1



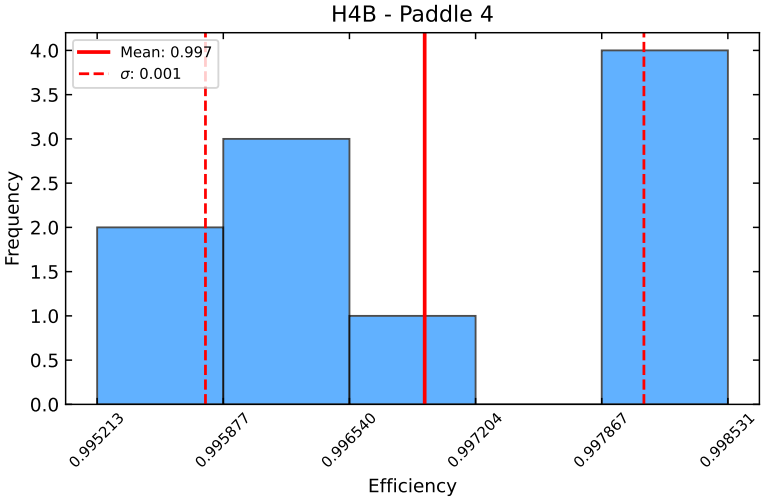
Efficiency Distribution: H4B - Paddle 2



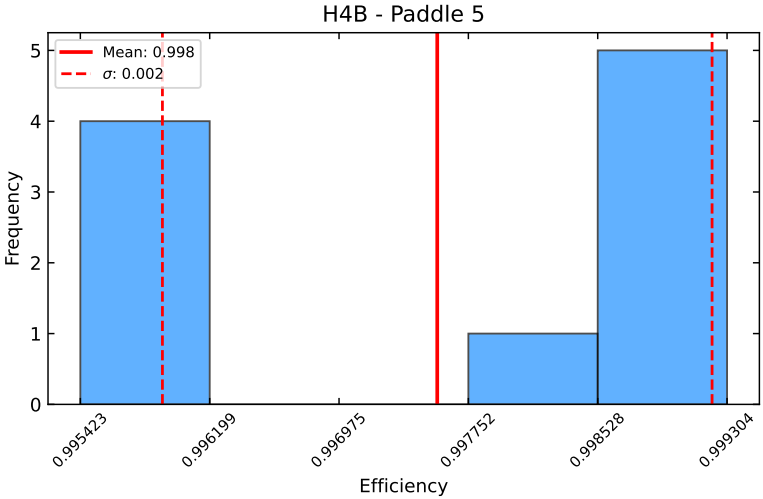
Efficiency Distribution: H4B - Paddle 3



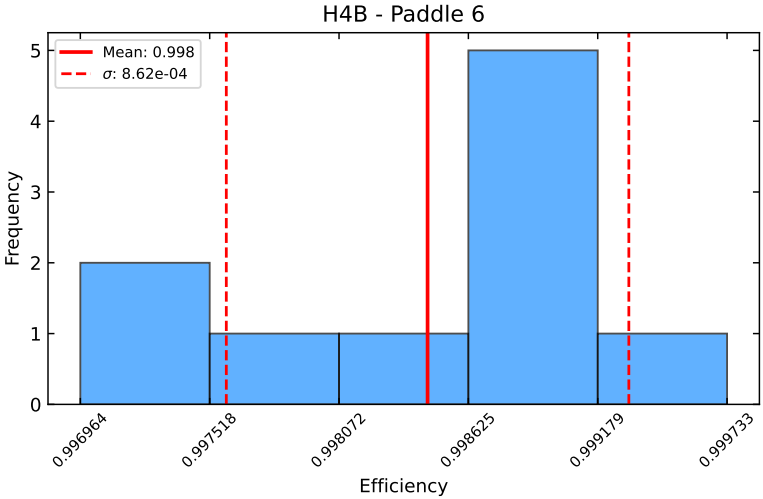
Efficiency Distribution: H4B - Paddle 4



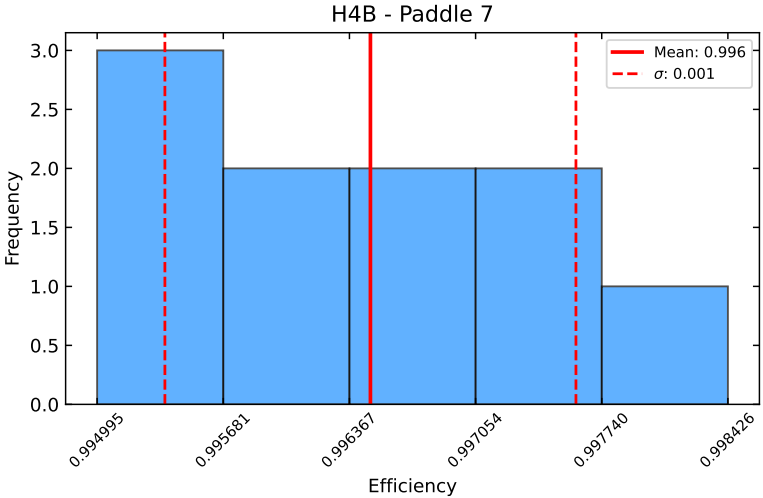
Efficiency Distribution: H4B - Paddle 5



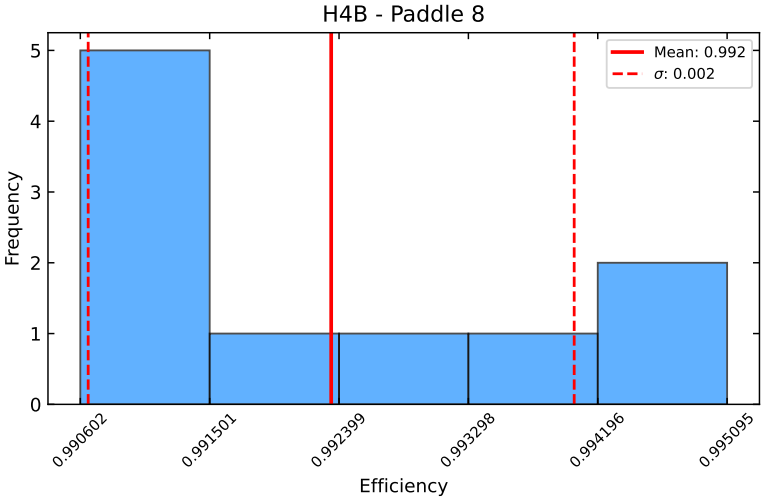
Efficiency Distribution: H4B - Paddle 6



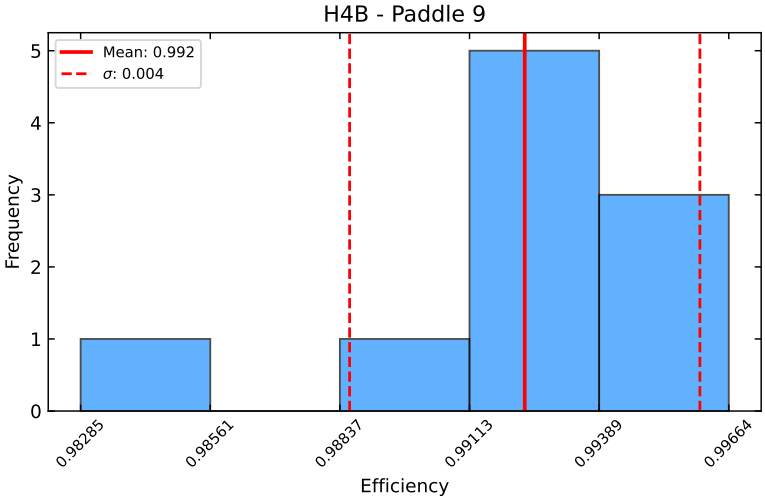
Efficiency Distribution: H4B - Paddle 7



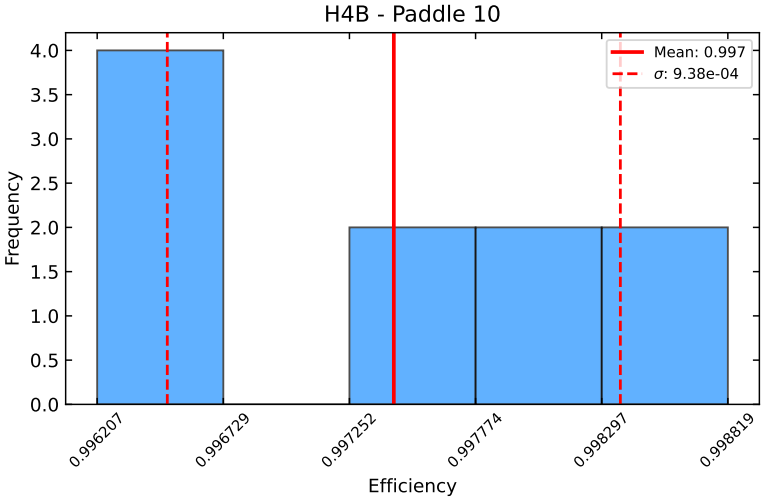
Efficiency Distribution: H4B - Paddle 8



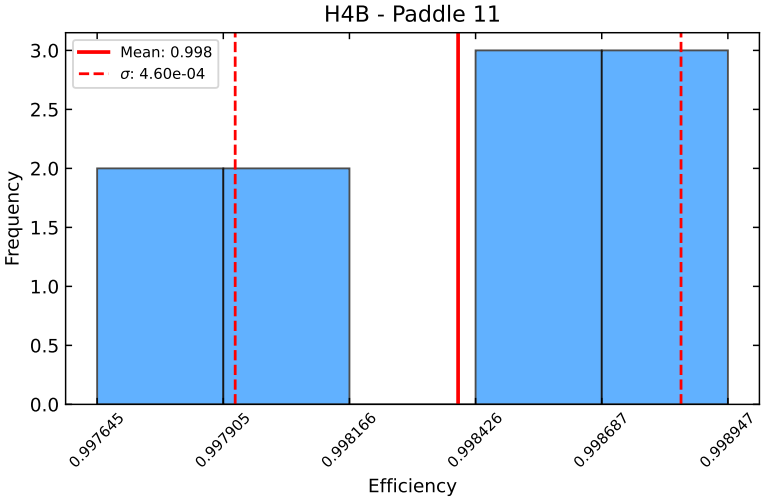
Efficiency Distribution: H4B - Paddle 9



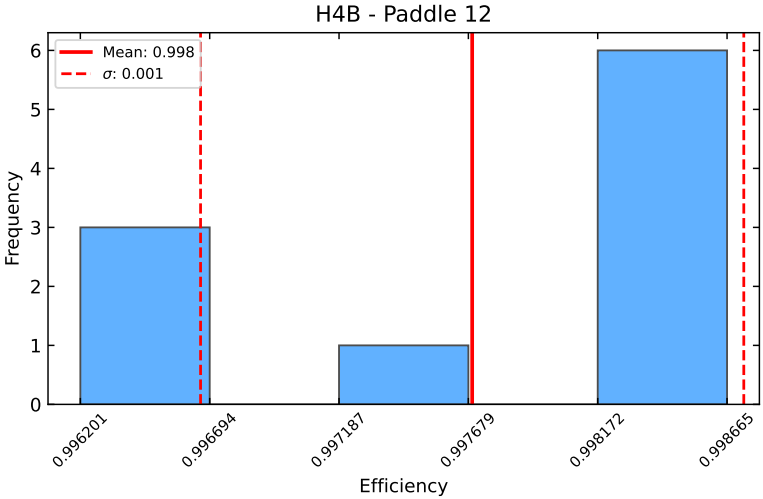
Efficiency Distribution: H4B - Paddle 10



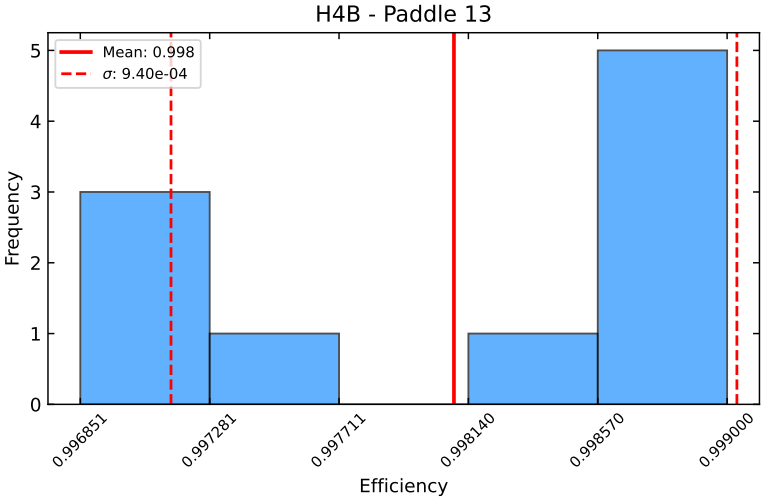
Efficiency Distribution: H4B - Paddle 11



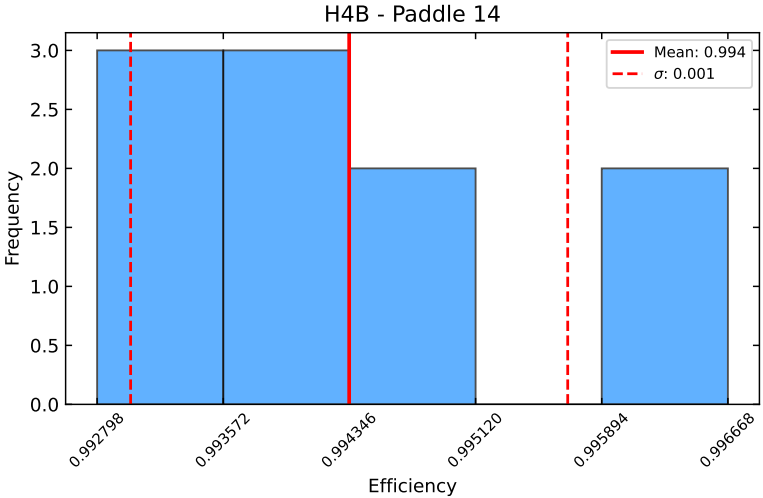
Efficiency Distribution: H4B - Paddle 12



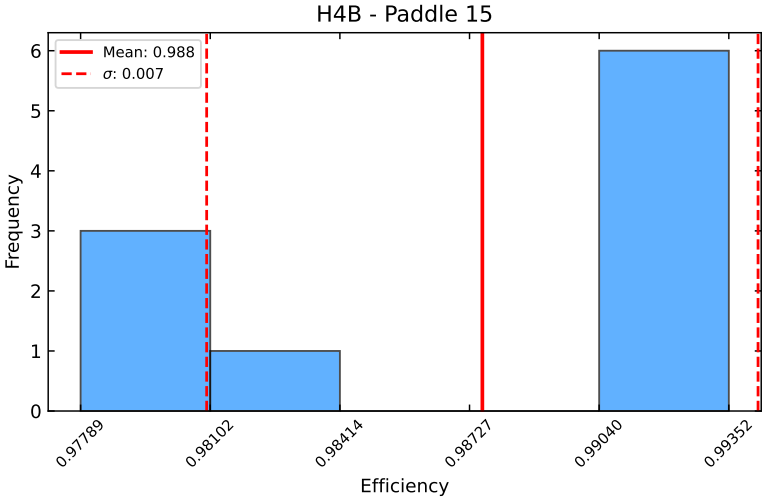
Efficiency Distribution: H4B - Paddle 13



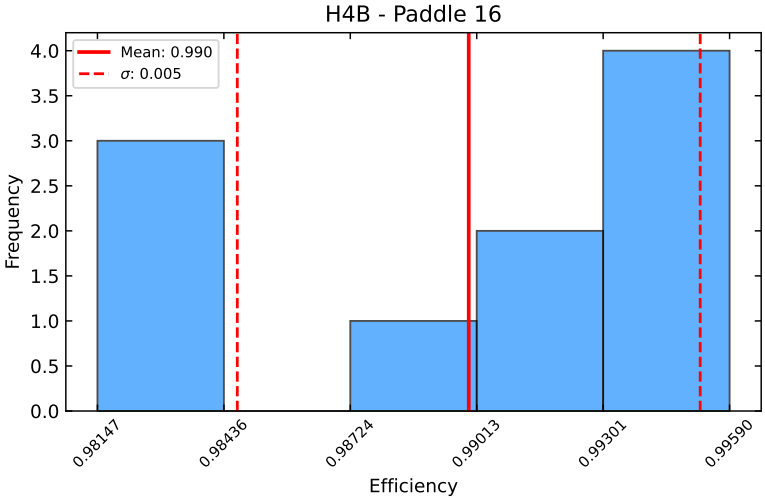
Efficiency Distribution: H4B - Paddle 14



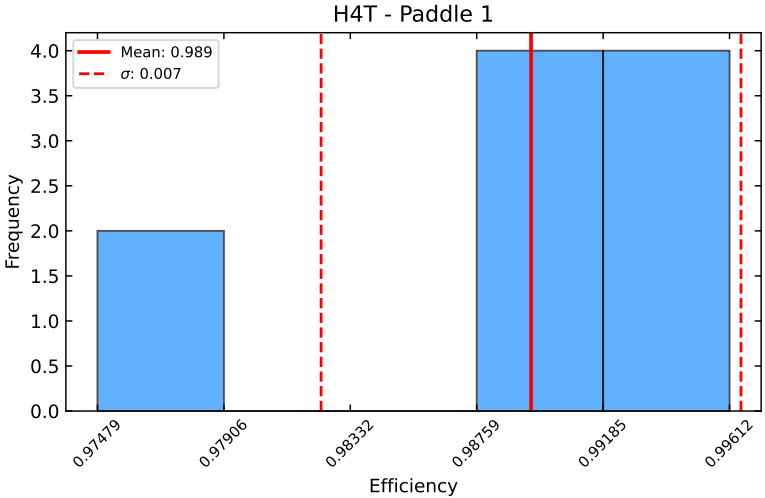
Efficiency Distribution: H4B - Paddle 15



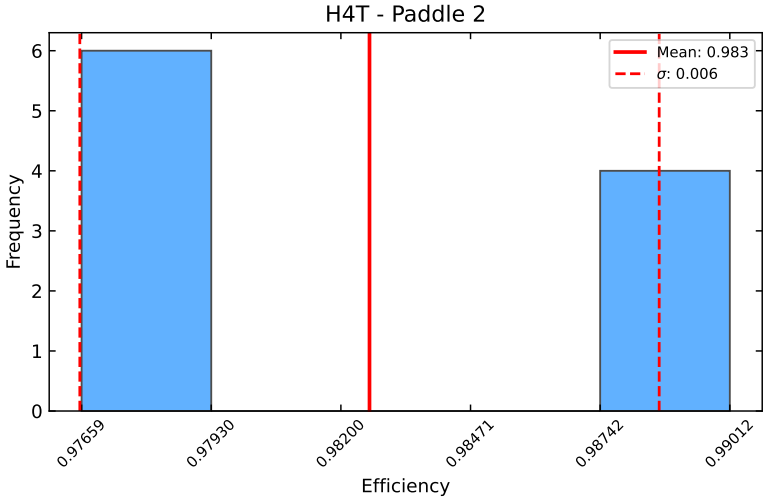
Efficiency Distribution: H4B - Paddle 16



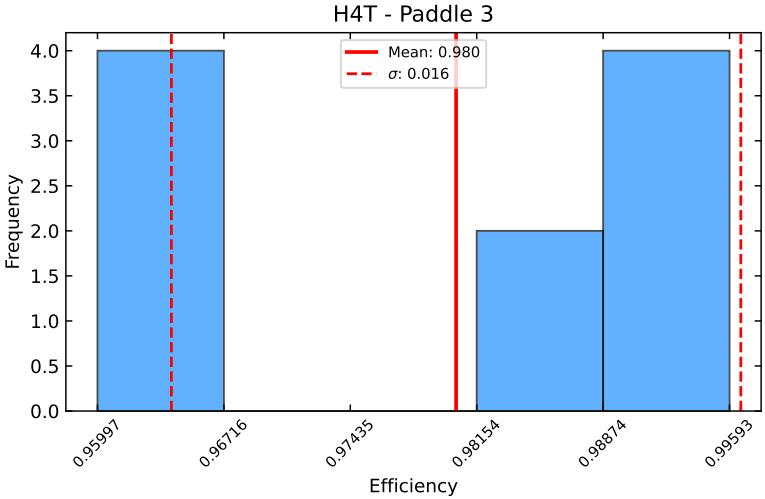
Efficiency Distribution: H4T - Paddle 1



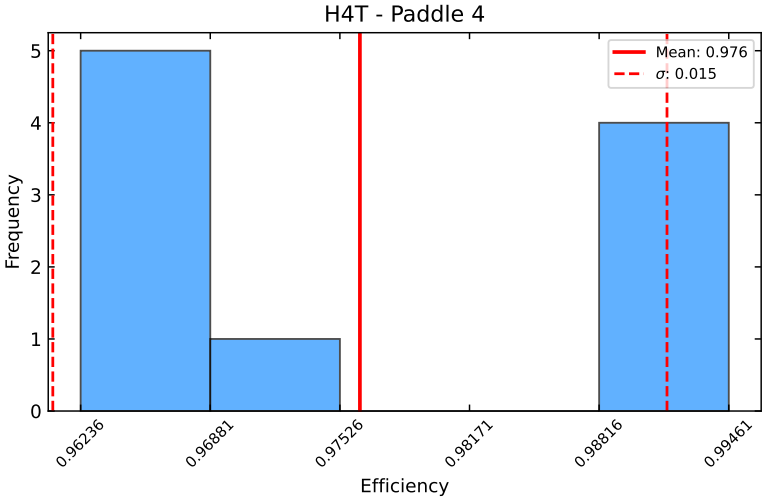
Efficiency Distribution: H4T - Paddle 2



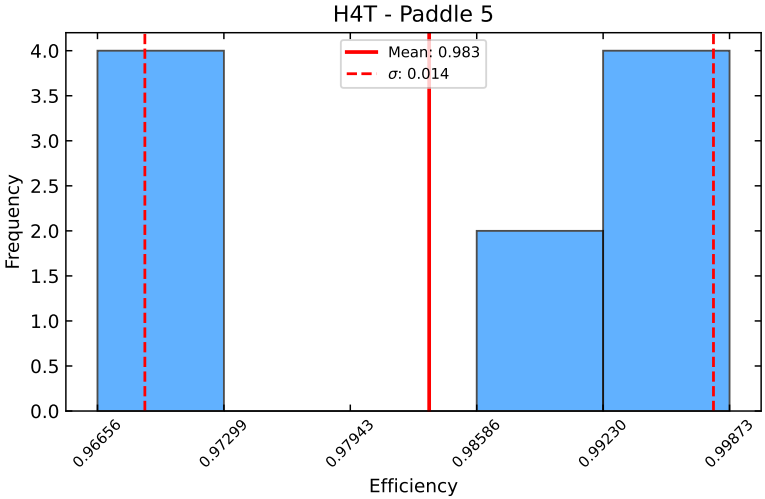
Efficiency Distribution: H4T - Paddle 3



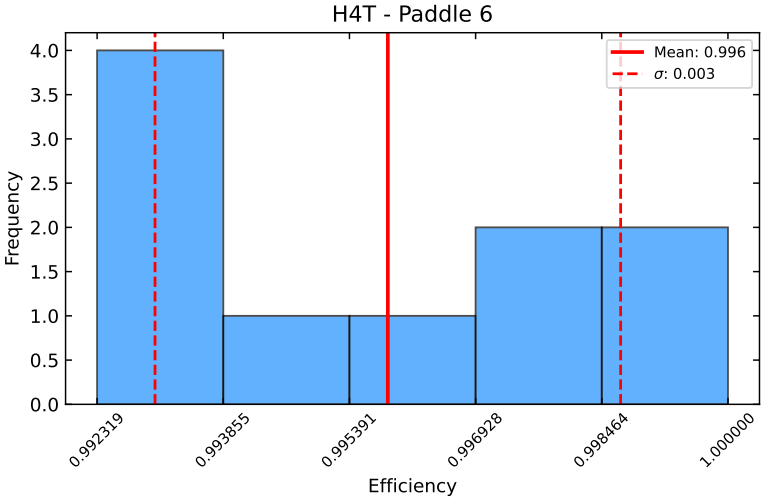
Efficiency Distribution: H4T - Paddle 4



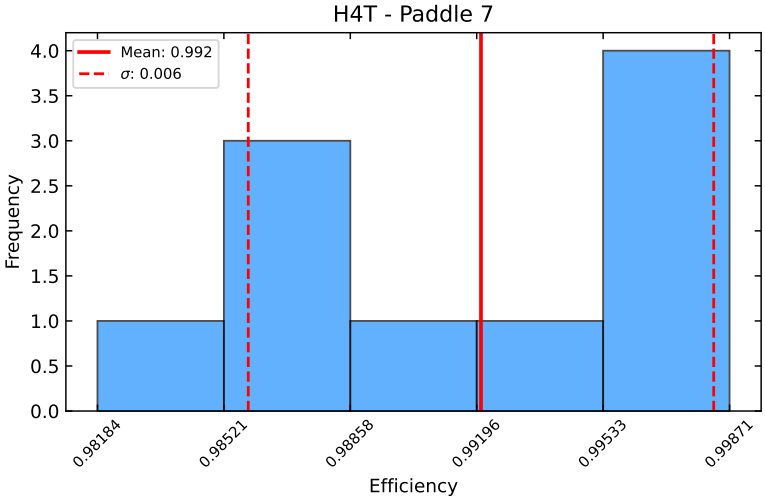
Efficiency Distribution: H4T - Paddle 5



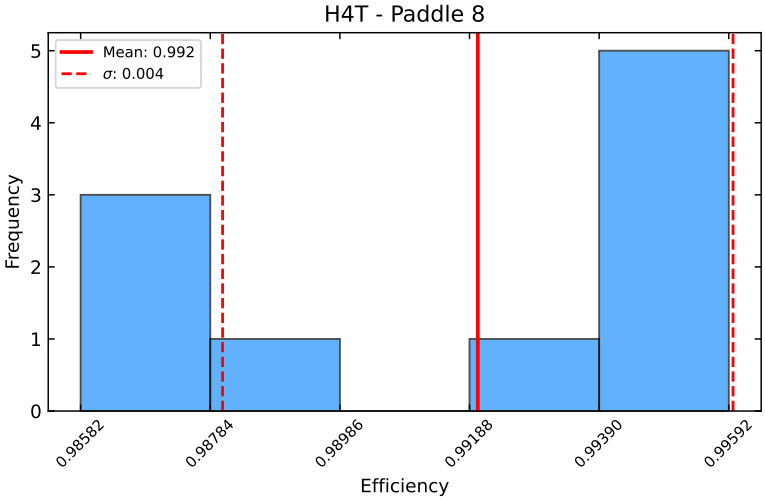
Efficiency Distribution: H4T - Paddle 6



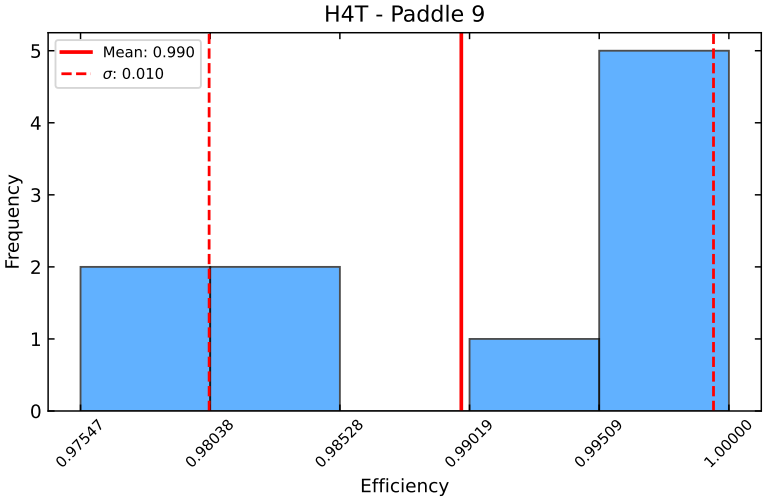
Efficiency Distribution: H4T - Paddle 7



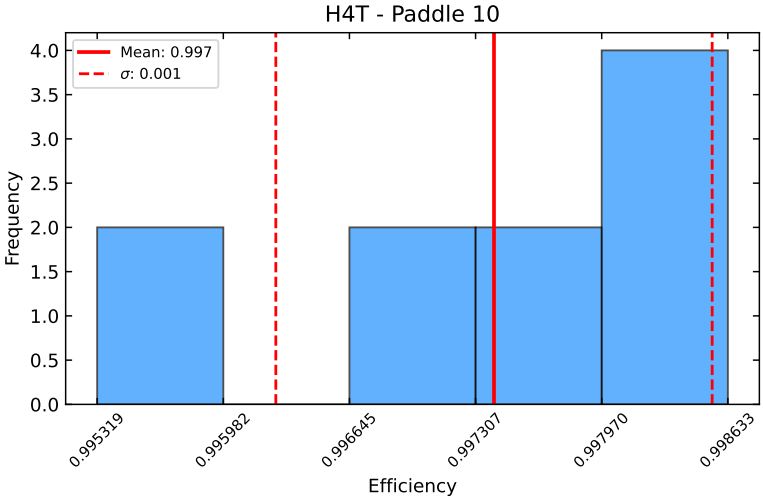
Efficiency Distribution: H4T - Paddle 8



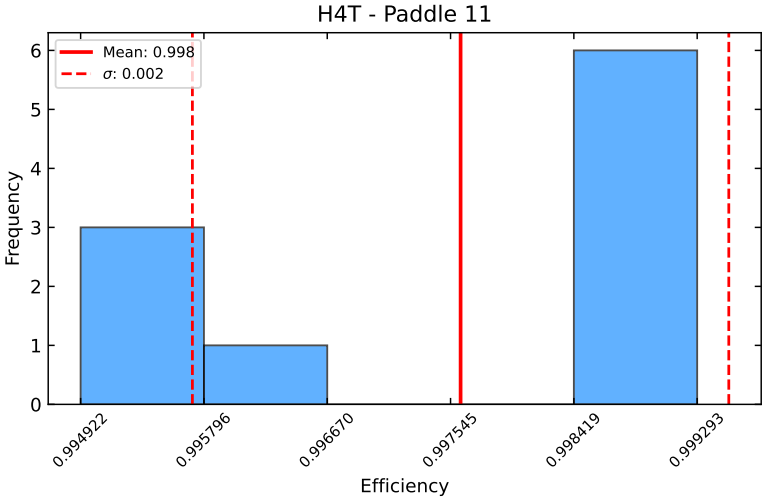
Efficiency Distribution: H4T - Paddle 9



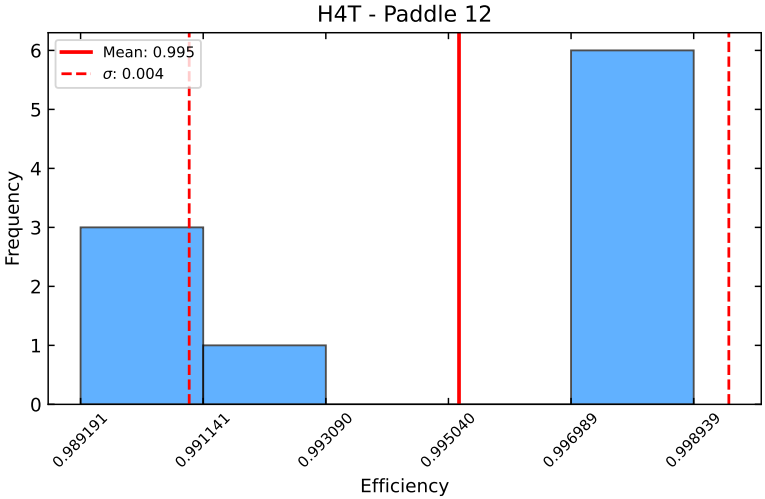
Efficiency Distribution: H4T - Paddle 10



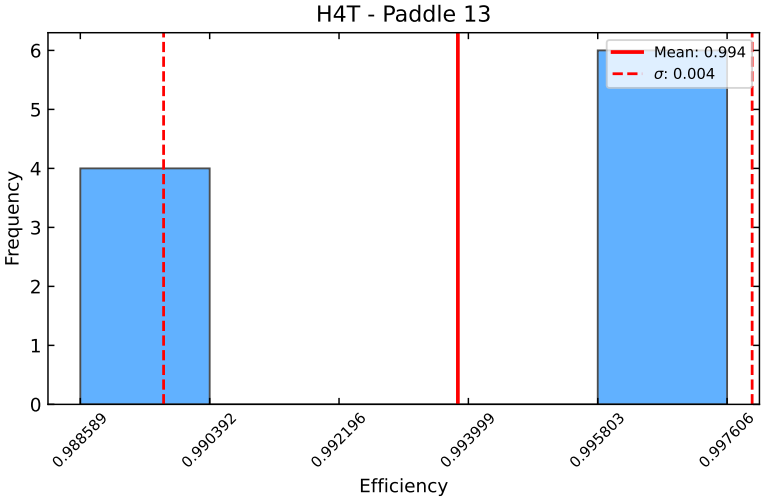
Efficiency Distribution: H4T - Paddle 11



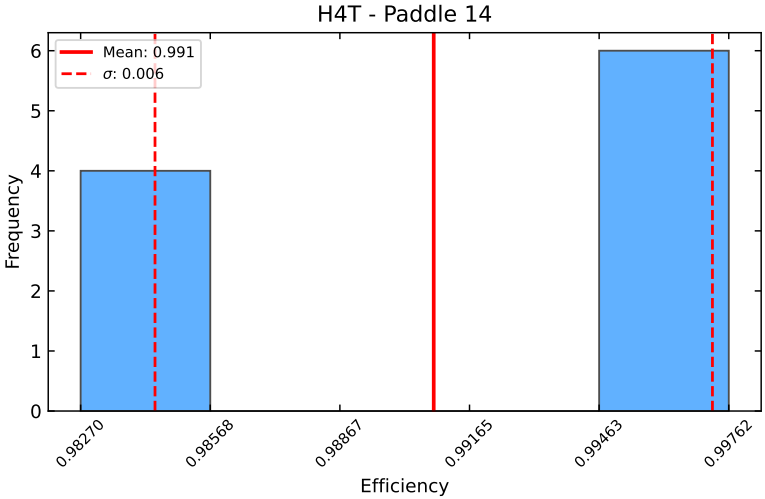
Efficiency Distribution: H4T - Paddle 12



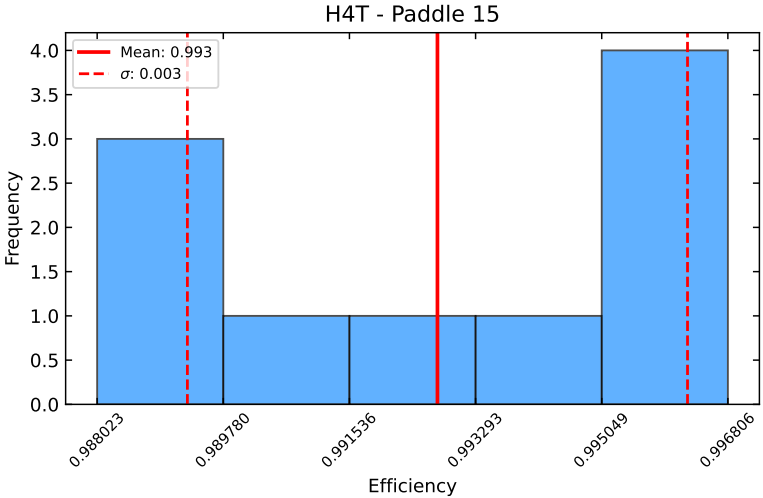
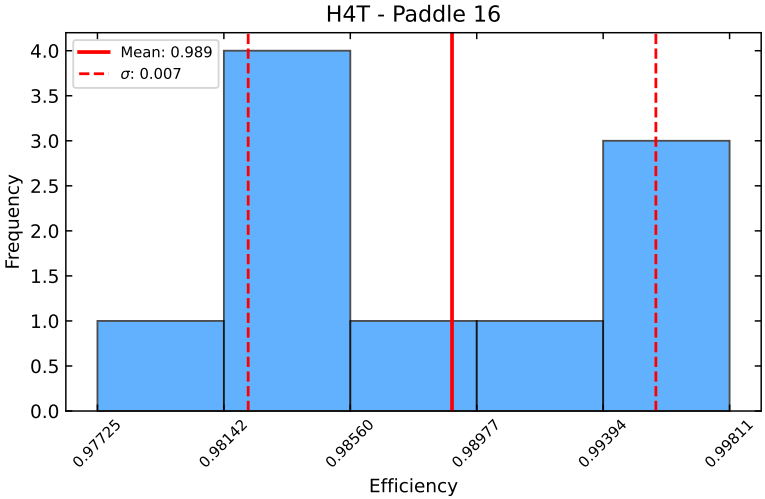
Efficiency Distribution: H4T - Paddle 13



Efficiency Distribution: H4T - Paddle 14



Efficiency Distribution: H4T - Paddle 15



Efficiency Distribution: H4T - Paddle 16